QuantiFi 2026 - Trade Tape & Market Participants

Who trades in the QuantiFi market, inferred from anonymous trades on days -2, -1, 0 for all 26 instruments · prepared 24 Oct 2026 following *quantifi_2026_trades_plan.md*.

Executive summary

- **Four participant types explain essentially the whole tape.** (1) A stateless market-maker bot that posts the visible book on every product; (2) a uniform-size noise taker on 15 of the 16 non-voucher products; (3) a rule-based volatility-arbitrage taker that dominates the voucher flow; (4) a jump-reactive momentum crowd on TELLURIUM. Counterparty IDs are blank, so these are behavioural inferences.
- **The resting side never reacts to trades.** After a print at the touch, the next snapshot shows the trade-depleted volume only 0-5% of the time, and the next mid moves with the trade on net -0.05 to +0.08 on 24 of 26 products. The book is regenerated each tick around a hidden fair value with fixed spreads.
- **Voucher takers follow an implied-volatility rule.** Their side has corr -0.47 to -0.87 with the strike's IV minus a fitted quadratic smile; 86-95% of prints at 7,000-7,900 are on the 'buy cheap / sell rich' side (98% when the residual is large). Rule-consistent prints earn +1.2 to +3.5 per lot at 100 ticks at 7,300-7,600: the only clearly informed flow in the market.
- **TELLURIUM flow is reactive, not informed.** 87% of its prints occur within 200 ticks after a jump; net flow in the 200 ticks before jumps is ~0, but +287 to +324 lots in the jump direction after. Takers chase, sweep deeper levels (23-34% of prints) and post-jump prints lose 35 per lot at 500 ticks.
- **Noise takers ignore every signal found in the EDA.** Their side is uncorrelated with the AMETHYST 10,000 anchor, the pearl-sum deviation that drives PINK_PEARLS, and the BLUE_PEARLS trend. Sizes are uniform on 1..N (N = 3, 4 or 6 by product), arrivals are Poisson-like, and they pay the half spread.
- **No 'buy-the-low' (Olivia-type) bot.** The share of prints within 50 ticks of each day's ex-post high/low is 0-4.7% against a 2% chance rate, and where it is elevated (TELLURIUM) takers are on the wrong side. No product has a distinct large-size print (maximum size 3-10).
- **No cross-product traders.** Co-timed prints across voucher strikes occur at exactly the independent-arrival rate with ~50% same-side; pearls and basket legs almost never print together.

Player map

Product	Players identified	Implication
AMETHYST	Market-maker bot + pure noise taker	Ideal counterparty. Quote both sides at or just inside the touch; no adverse selection to price in.
PAPER_LANTERNS	Market-maker bot + pure noise taker	Same as AMETHYST: flow is safe to make against; price noise, not flow, sets the edge.
TOFFEE_CAKE	Market-maker bot + noise taker with clustered arrivals	Making against this flow earns more than the spread on two of three days.
GOLDEN_PEARLS	Market-maker bot + noise taker	Flow is benign; the risk is the steps themselves, not the counterparties.
PINK_PEARLS	Market-maker bot + noise taker	The pearl-sum edge from the EDA report is uncontested by other takers.
BLUE_PEARLS	Market-maker bot + noise taker	Lean passive quotes with the trend; the flow will not compete for it.
WHITE_PEARLS	Market-maker bot + noise taker	Benign flow; the 20-lot limit is the binding constraint, not adverse selection.
BLACK_PEARLS	Market-maker bot + noise taker	Among pearls, the best product to provide liquidity in.
JELLY	Market-maker bot + noise taker	Spread capture is real; manage inventory, not counterparty risk.
PEANUT_BUTTER	Market-maker bot + noise taker	Benign flow in a volatile product; inventory risk dominates.
LYCHEE	Market-maker bot + noise taker	Safe to make and to use as the voucher hedge; voucher takers do not trade LYCHEE momentum.
DRAGONFRUIT	Market-maker bot + noise taker (thin)	Too little flow to identify a player beyond noise; treat as benign.
GRANADILLA	Market-maker bot + noise taker (thin)	Benign but thin.
MANGOSTEEN	Market-maker bot + noise taker (thin)	Benign but thin; statistics are low-power.
FRUIT_BASKET	Market-maker bot + noise taker (thin)	Nobody arbitrages the basket in the tape; a basket strategy competes only with the bot's quotes.
TELLURIUM	Market-maker bot + jump-reactive momentum takers (fast and sweeping) + background noise	Pull or widen quotes for ~200 ticks after a jump; the chasers' one-sided flow runs over resting orders.
VOUCHER_6800..8100	Market-maker bot + IV-smile value taker (~90% of flow) + residual noise	Do not be the passive side to rule-consistent flow; trade the IV residual first

Method

Each print is matched to the same-timestamp book (100% match; every print is at or through the touch). Aggressor side uses the quote rule, which is exact here because no print is inside the spread, so the Lee & Ready (1991) tick-test fallback is never needed. For each product and day we measure: size distribution (chi-square vs uniform), price level hit and multi-level sweeps; dispersion of 100-tick trade counts and inter-arrival CV; trade-sign autocorrelation and runs; correlation of side with real-time state (mid minus EWMA-100, 20-tick momentum, position in the running daily range); aggressor markouts at 1/5/20/100/500 ticks; the effective half spread split into realized spread and price impact (Huang & Stoll 1997; Hendershott, Jones & Menkveld 2011); a Hasbrouck (1991) tick-time VAR with 5 lags for the cumulative impact of a buy; and EKOP PIN on 50 buckets per day. A Gaussian mixture on product-normalised ex-ante features groups prints into archetypes; markouts are used only to validate the groups.

Participant taxonomy tested

Archetype	Literature	Footprint tested	Found?
Liquidity provider / dealer	Harris (2003) ch. 13-14; Ho & Stoll (1981)	Fixed spread set, L1 volume range, response to trades	Yes: one stateless bot on every product
Noise / liquidity trader	Kyle (1985); Glosten & Milgrom (1985)	Uniform size, Poisson arrival, random side, markout = -half spread	Yes: 15 of 16 non-voucher products
Informed / value trader	Glosten & Milgrom (1985); Harris (2003) ch. 10	Side vs fair-value deviation, positive long-horizon markout	Yes: vouchers (IV-smile value rule)
Momentum / trend chaser	Harris (2003); Bouchaud et al. (2018)	Side vs past return, flow after moves, sweeps	Yes: TELLURIUM, triggered by jumps
Order splitter / metaorder	Lillo & Farmer (2004)	Trade-sign long memory, runs	Apparent in vouchers/TELLURIUM; explained by persistent signals
Arbitrageur / hedger	Harris (2003) ch. 17	Excess co-timed trades with linked sides	No: co-timing equals independence
Extreme-timing bot ('Olivia')	Frankfurt Hedgehogs Prosperity-3 write-up	Prints at ex-post daily high/low, distinct size	No

Tape audit (T0)

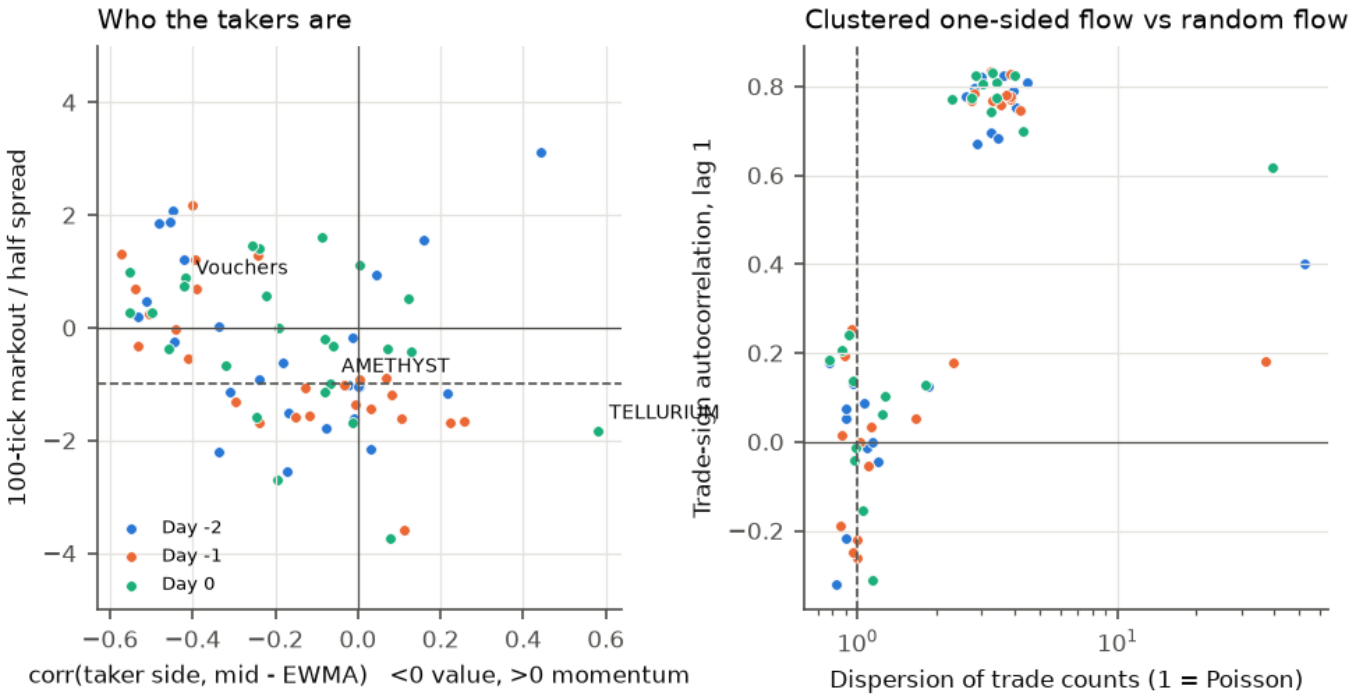
Product	Prints/day	Lots/day	Sizes	Uniform p (min)	L2+ sweeps	Lots / limit
AMETHYST	135-159	323-396	1-4	0.249	0%	6.5-7.9x
BLACK_PEARLS	51-58	91-112	1-3	0.113	0%	4.5-5.6x
BLUE_PEARLS	50-63	94-130	1-3	0.214	0%	4.7-6.5x
DRAGONFRUIT	15-23	28-48	1-3	0.438	0%	0.7-1.2x
FRUIT_BASKET	15-17	32-37	1-3	0.074	0%	0.8-0.9x
GOLDEN_PEARLS	49-66	98-130	1-3	0.271	0%	4.9-6.5x
GRANADILLA	19-33	41-66	1-3	0.104	0%	1.0-1.6x
JELLY	59-68	156-171	1-4	0.057	0%	3.1-3.4x
LYCHEE	53-63	119-168	1-4	0.196	0%	0.6-0.8x
MANGOSTEEN	12-23	21-49	1-3	0.032	0%	0.5-1.2x
PAPER_LANTERNS	101-104	252-264	1-4	0.478	0%	5.0-5.3x
PEANUT_BUTTER	58-79	148-194	1-4	0.214	0%	3.0-3.9x
PINK_PEARLS	61-62	126-134	1-3	0.230	0%	6.3-6.7x
TELLURIUM	899-1600	3875-6631	1-10	0.000	28%	96.9-165.8x
TOFFEE_CAKE	41-48	148-167	1-6	0.102	0%	3.0-3.3x
VOUCHER_6800	477-604	993-1201	1-3	0.052	0%	19.9-24.0x
VOUCHER_7000	520-659	1032-1349	1-3	0.210	0%	20.6-27.0x
VOUCHER_7200	589-664	1173-1340	1-3	0.067	0%	23.5-26.8x
VOUCHER_7300	587-639	1186-1287	1-3	0.765	0%	23.7-25.7x
VOUCHER_7400	508-557	1014-1107	1-3	0.404	0%	20.3-22.1x
VOUCHER_7500	480-656	957-1315	1-3	0.161	0%	19.1-26.3x
VOUCHER_7600	529-591	1043-1158	1-3	0.278	0%	20.9-23.2x
VOUCHER_7750	524-642	1050-1291	1-3	0.720	0%	21.0-25.8x
VOUCHER_7900	547-604	1127-1249	1-3	0.033	0%	22.5-25.0x
VOUCHER_8100	632-669	1248-1338	1-3	0.048	0%	25.0-26.8x
WHITE_PEARLS	61-67	108-138	1-3	0.069	0%	5.4-6.9x

Player 1: the market-maker bot (passive side)

- Spreads take a small fixed set of values per product (a single value of 3 for VOUCHER_7200-8100; 3-5 values for most others; 10 for TOFFEE_CAKE and 14 for TELLURIUM, whose quotes widen around jumps).
- Level-1 sizes range 5-20/25/30 (LYCHEE 10-40; TELLURIUM 2-24) and do not respond to trades: after a print at the touch, the next snapshot shows 'previous volume minus traded size' only 0-5% of the time.
- Trades do not move the bot's mid: the net share of next-tick mid moves in the trade's direction is -0.05 to +0.08 on 24 products (FRUIT_BASKET +0.21 on 47 prints; TELLURIUM +0.72, driven by post-jump momentum).
- Implication: the passive side is a quote generator around a hidden fair value, not a strategic dealer with inventory. Our own quotes compete only with this bot, whose levels are predictable from the book.

Product	Spread mode (# values)	L1 size range	Depleted after trade	Next mid with trade (net)
AMETHYST	3 (3)	5-30	1.4%	+0.03
BLACK_PEARLS	8 (3)	5-25	0.6%	+0.01
BLUE_PEARLS	6 (4)	5-25	0.0%	+0.01
DRAGONFRUIT	4 (3)	5-20	0.0%	-0.05
FRUIT_BASKET	6 (3)	5-20	0.0%	+0.21
GOLDEN_PEARLS	3 (3)	5-25	1.1%	-0.01
GRANADILLA	4 (3)	5-20	2.7%	+0.08
JELLY	11 (5)	5-25	0.5%	-0.01
LYCHEE	4 (4)	10-40	1.7%	+0.01
MANGOSTEEN	4 (3)	5-20	0.0%	-0.05
PAPER_LANTERNS	4 (3)	5-30	1.0%	+0.01
PEANUT_BUTTER	6 (3)	5-25	0.0%	+0.07
PINK_PEARLS	4 (4)	5-25	2.7%	+0.05
TELLURIUM	9 (14)	2-24	0.5%	+0.72
TOFFEE_CAKE	15 (10)	5-30	1.5%	+0.00
VOUCHER_6800	6 (3)	5-20	1.1%	+0.02
VOUCHER_7000	6 (3)	5-20	0.9%	+0.00
VOUCHER_7200	3 (1)	5-20	1.7%	+0.03
VOUCHER_7300	3 (1)	5-20	2.4%	+0.02
VOUCHER_7400	3 (1)	5-20	2.9%	+0.04
VOUCHER_7500	3 (1)	5-20	1.9%	+0.05
VOUCHER_7600	3 (1)	5-20	2.9%	+0.05
VOUCHER_7750	3 (1)	5-20	3.3%	+0.03
VOUCHER_7900	3 (1)	5-20	4.3%	+0.03
VOUCHER_8100	3 (1)	5-20	5.3%	+0.01
WHITE_PEARLS	4 (3)	5-25	1.5%	+0.08

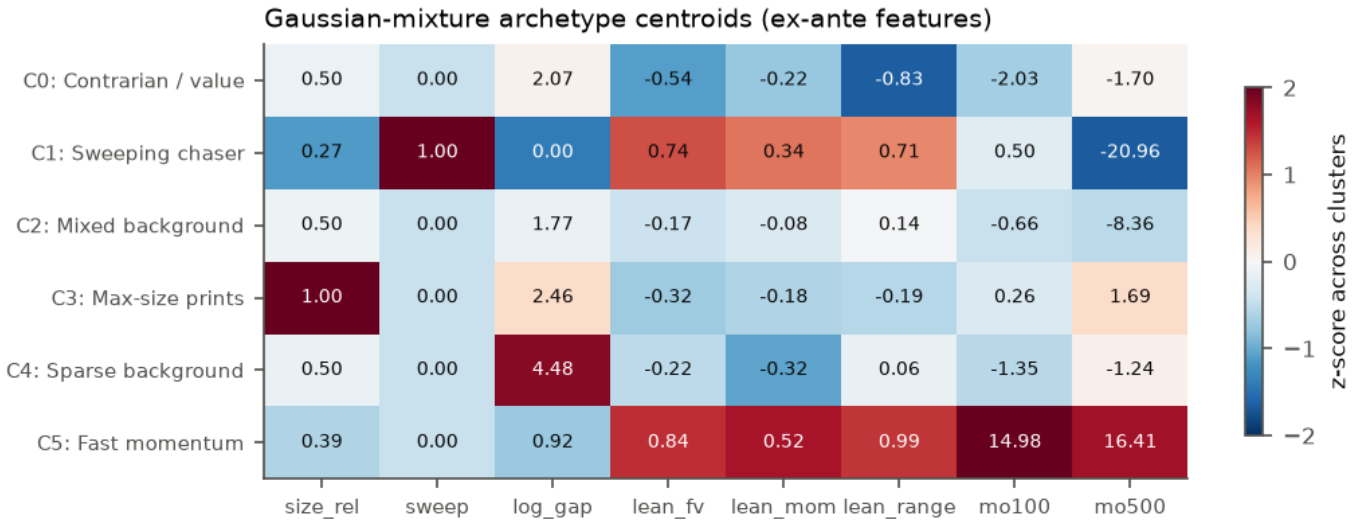
Cross-product evidence



Information content of takers by day (T4)

Product	Markout h1 (days -2/-1/0)	Markout h100	Realized spread h100	Hasbrouck LR / half spread	PIN
AMETHYST	-1.5 / -1.5 / -1.5	-1.5 / -1.5 / -1.5	+1.5 / +1.5 / +1.5	-0.01 / +0.02 / -0.02	0.00 / 0.48 / 0.31
BLACK_PEARLS	-3.6 / -4.1 / -4.1	-5.9 / -5.7 / -6.3	+5.9 / +5.8 / +6.3	+0.10 / -0.10 / +0.23	0.38 / 0.14 / 0.77
BLUE_PEARLS	-2.9 / -3.0 / -2.8	-5.4 / -2.8 / -2.0	+5.4 / +2.8 / +2.0	+0.14 / +0.17 / +0.10	0.00 / 0.55 / 0.62
DRAGONFRUIT	-2.5 / -2.2 / -2.2	-2.4 / -2.6 / -5.6	+2.4 / +2.6 / +5.6	+0.31 / -0.14 / -0.27	0.00 / 0.45 / 0.00
FRUIT_BASKET	-2.2 / -4.2 / -2.9	+23.8 / -16.1 / +4.1	-23.8 / +16.1 / -4.1	-1.47 / -0.68 / -0.81	0.00 / 1.00 / 0.00
GOLDEN_PEARLS	-1.4 / -1.6 / -1.6	-1.5 / -2.5 / -0.6	+1.5 / +2.4 / +0.6	-0.02 / -0.07 / +0.12	0.24 / 0.66 / 0.06
GRANADILLA	-1.8 / -1.8 / -2.1	-1.3 / -3.3 / -7.7	+1.3 / +3.3 / +7.7	-0.27 / +0.32 / -0.52	0.35 / 0.00 / 0.00
JELLY	-5.8 / -5.7 / -5.0	-1.0 / -6.5 / +2.9	+1.0 / +6.5 / -2.9	+0.11 / +0.03 / +0.17	0.54 / 0.00 / 0.00
LYCHEE	-1.8 / -2.1 / -1.9	-4.4 / -2.2 / -3.3	+4.4 / +2.2 / +3.3	+0.16 / -0.16 / -0.05	0.69 / 0.24 / 0.00
MANGOSTEEN	-2.6 / -1.8 / -2.5	+3.0 / +2.6 / -17.6	-3.0 / -2.6 / +17.6	-0.68 / -0.44 / -0.20	0.00 / 0.39 / 0.00
PAPER_LANTERNS	-2.0 / -2.0 / -1.9	-2.1 / -1.8 / -2.3	+2.1 / +1.8 / +2.3	+0.11 / +0.03 / -0.00	0.00 / 0.00 / 0.00
PEANUT_BUTTER	-2.5 / -2.7 / -3.9	+2.8 / -4.0 / +0.0	-2.8 / +4.0 / +0.0	+0.23 / -0.24 / -0.43	0.00 / 0.14 / 0.03
PINK_PEARLS	-1.9 / -2.1 / -1.9	-3.1 / -3.2 / -0.7	+3.1 / +3.3 / +0.7	+0.14 / -0.09 / -0.03	0.57 / 0.00 / 0.61
TELLURIUM	-5.2 / -4.6 / -4.6	+25.2 / -11.3 / -13.3	-25.2 / +11.3 / +13.3	+1.46 / +1.08 / +1.02	0.83 / 0.80 / 0.74
TOFFEE_CAKE	-7.4 / -7.0 / -6.9	-18.2 / -11.9 / -1.4	+18.0 / +11.9 / +1.4	-0.34 / +0.04 / -0.09	0.35 / 0.52 / 0.31
VOUCHER_6800	-3.0 / -3.0 / -3.0	-3.4 / -5.0 / -1.0	+3.4 / +5.0 / +1.0	+0.02 / -0.03 / +0.05	0.67 / 0.74 / 0.72
VOUCHER_7000	-2.9 / -3.1 / -3.0	-2.7 / -1.6 / +3.4	+2.7 / +1.6 / -3.4	+0.01 / -0.01 / +0.07	0.78 / 0.79 / 0.79
VOUCHER_7200	-1.5 / -1.5 / -1.4	+0.0 / -0.0 / +0.8	-0.0 / +0.0 / -0.8	+0.15 / +0.04 / +0.09	0.77 / 0.81 / 0.85
VOUCHER_7300	-1.4 / -1.5 / -1.5	+2.8 / +1.0 / +2.2	-2.8 / -1.0 / -2.2	+0.24 / +0.05 / +0.01	0.81 / 0.79 / 0.79
VOUCHER_7400	-1.5 / -1.5 / -1.4	+1.8 / +3.3 / +1.3	-1.8 / -3.3 / -1.3	+0.06 / +0.11 / +0.15	0.71 / 0.63 / 0.64
VOUCHER_7500	-1.4 / -1.4 / -1.5	+3.1 / +1.8 / +1.1	-3.1 / -1.8 / -1.1	+0.29 / +0.12 / +0.07	0.77 / 0.78 / 0.75
VOUCHER_7600	-1.4 / -1.5 / -1.5	+2.8 / +1.0 / +1.5	-2.8 / -1.0 / -1.5	+0.19 / +0.10 / +0.06	0.62 / 0.57 / 0.67
VOUCHER_7750	-1.5 / -1.5 / -1.5	+0.3 / +2.0 / +0.4	-0.3 / -2.0 / -0.4	+0.06 / +0.07 / +0.08	0.77 / 0.72 / 0.78
VOUCHER_7900	-1.5 / -1.5 / -1.5	+0.7 / +0.4 / +0.4	-0.7 / -0.4 / -0.4	+0.08 / +0.06 / +0.05	0.85 / 0.74 / 0.76
VOUCHER_8100	-1.5 / -1.5 / -1.5	-0.4 / -0.5 / -0.6	+0.4 / +0.5 / +0.6	+0.03 / +0.03 / +0.02	0.76 / 0.72 / 0.82
WHITE_PEARLS	-1.6 / -2.0 / -1.8	-4.3 / -7.4 / +3.2	+4.3 / +7.4 / -3.2	+0.14 / +0.65 / -0.10	0.00 / 0.06 / 0.06

Archetype clustering (T8)



- BIC keeps improving up to the 6-component cap, so k = 6 is a readability choice, not a statistical optimum.
- Two clusters are TELLURIUM-specific and behaviourally distinct (fast momentum vs sweeping chaser); their validation markouts differ sharply (+15 at 100 ticks vs -21 at 500 ticks).
- 'Contrarian / value' collects many rule-consistent voucher prints near daily range extremes.
- 'Max-size prints', 'Sparse background' and 'Mixed background' split the uniform-size noise population by size and gap. Their markouts are similar, so they are one player type seen from different angles - a known artefact of mixture models on uniform features.

Archetype mix by product and day

Product	Day	C0 Contrarian / value	C1 Sweeping chaser	C2 Mixed background	C3 Max-size prints	C4 Sparse background	C5 Fast momentum
AMETHYST	-2	12%	0%	12%	26%	49%	1%
	-1	17%	0%	15%	22%	45%	1%
	0	10%	0%	18%	23%	48%	1%
BLACK_PEARLS	-2	5%	0%	13%	25%	56%	0%
	-1	6%	0%	6%	25%	63%	0%
	0	7%	0%	5%	29%	59%	0%
BLUE_PEARLS	-2	6%	0%	6%	30%	58%	0%
	-1	6%	0%	2%	38%	54%	0%
	0	17%	0%	3%	34%	46%	0%
DRAGONFRUIT	-2	0%	0%	4%	43%	52%	0%
	-1	0%	0%	0%	45%	55%	0%
	0	0%	0%	7%	27%	67%	0%
FRUIT_BASKET	-2	0%	0%	0%	33%	67%	0%
	-1	6%	0%	0%	41%	53%	0%
	0	0%	0%	0%	60%	40%	0%
GOLDEN_PEARLS	-2	12%	0%	6%	39%	43%	0%
	-1	6%	0%	3%	27%	61%	2%
	0	17%	0%	5%	35%	44%	0%
GRANADILLA	-2	13%	0%	0%	39%	48%	0%
	-1	3%	0%	0%	30%	67%	0%
	0	0%	0%	5%	37%	58%	0%
JELLY	-2	3%	0%	8%	34%	54%	0%
	-1	7%	0%	7%	34%	51%	0%
	0	6%	0%	2%	24%	67%	2%
LYCHEE	-2	6%	0%	4%	21%	70%	0%
	-1	3%	0%	8%	29%	60%	0%
	0	5%	0%	9%	37%	49%	0%
MANGOSTEEN	-2	0%	0%	0%	25%	75%	0%
	-1	4%	0%	0%	52%	43%	0%

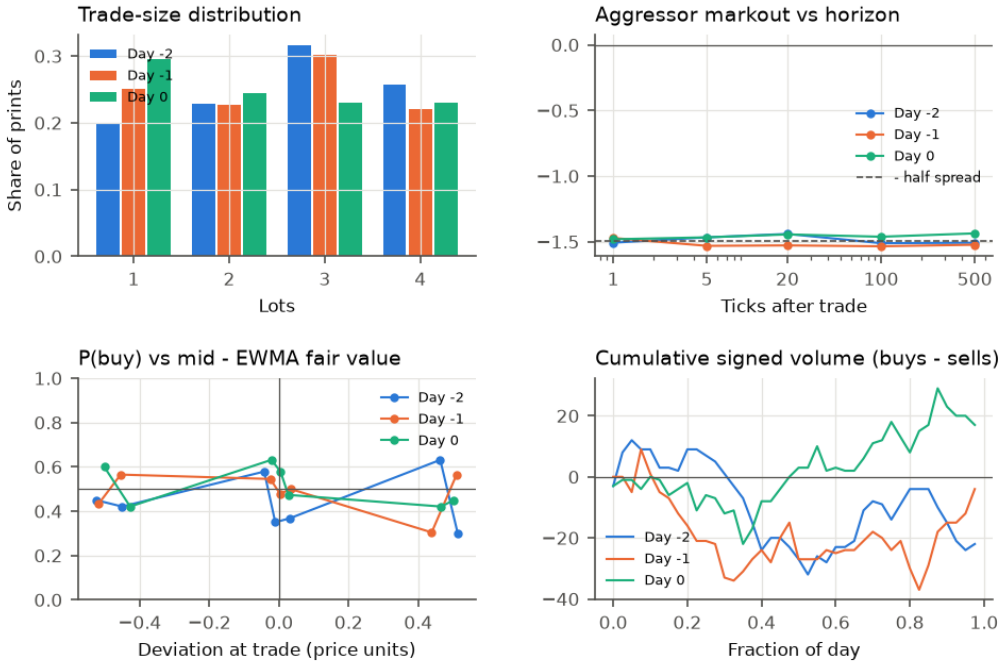
Product	Day	C0 Contrarian / value	C1 Sweeping chaser	C2 Mixed background	C3 Max-size prints	C4 Sparse background	C5 Fast momentum
	0	0%	0%	0%	36%	64%	0%
PAPER_LANTERNS	-2	11%	0%	12%	25%	53%	0%
	-1	21%	0%	10%	22%	48%	0%
	0	17%	0%	4%	21%	57%	0%
PEANUT_BUTTER	-2	9%	0%	11%	23%	57%	0%
	-1	10%	0%	10%	34%	45%	0%
	0	2%	0%	8%	24%	66%	0%
PINK_PEARLS	-2	7%	0%	5%	39%	49%	0%
	-1	15%	0%	3%	39%	43%	0%
	0	11%	0%	3%	44%	42%	0%
TELLURIUM	-2	8%	34%	14%	1%	4%	39%
	-1	9%	23%	29%	3%	5%	31%
	0	12%	27%	33%	2%	4%	22%
TOFFEE_CAKE	-2	2%	0%	32%	17%	49%	0%
	-1	6%	0%	25%	17%	50%	2%
	0	0%	0%	22%	11%	67%	0%
VOUCHER_6800	-2	16%	0%	34%	39%	12%	0%
	-1	19%	0%	37%	34%	11%	0%
	0	9%	0%	47%	34%	10%	0%
VOUCHER_7000	-2	20%	0%	36%	37%	7%	1%
	-1	20%	0%	38%	32%	9%	0%
	0	13%	0%	43%	34%	11%	0%
VOUCHER_7200	-2	20%	0%	38%	33%	9%	0%
	-1	22%	0%	38%	31%	8%	0%
	0	10%	0%	48%	33%	10%	0%
VOUCHER_7300	-2	25%	0%	34%	33%	8%	0%
	-1	20%	0%	39%	35%	7%	0%
	0	16%	0%	41%	34%	9%	0%
VOUCHER_7400	-2	21%	0%	32%	34%	13%	0%
	-1	21%	0%	37%	32%	10%	1%
	0	10%	0%	46%	32%	12%	0%
VOUCHER_7500	-2	18%	0%	42%	31%	10%	0%
	-1	24%	0%	31%	35%	10%	0%
	0	20%	0%	38%	33%	10%	0%
VOUCHER_7600	-2	31%	0%	26%	33%	10%	0%
	-1	35%	0%	26%	30%	9%	0%
	0	22%	0%	35%	33%	9%	0%
VOUCHER_7750	-2	41%	0%	17%	34%	7%	0%
	-1	17%	0%	38%	33%	12%	0%
	0	27%	0%	31%	34%	8%	0%
VOUCHER_7900	-2	25%	0%	31%	37%	7%	0%
	-1	12%	0%	40%	38%	9%	1%
	0	18%	0%	36%	34%	12%	0%
VOUCHER_8100	-2	39%	0%	19%	36%	5%	0%
	-1	30%	0%	31%	31%	7%	0%
	0	35%	0%	23%	36%	6%	0%
WHITE_PEARLS	-2	13%	0%	3%	37%	46%	0%
	-1	13%	0%	3%	20%	64%	0%
	0	3%	0%	9%	30%	58%	0%

Cross-product co-timing (T7)

Family	Day	Trade ticks	Multi-product share	Expected if independent	Same-side share
Vouchers	-2	4528	23.2%	24.5%	49%
Vouchers	-1	4435	23.9%	24.2%	53%
Vouchers	0	4675	26.0%	25.9%	45%
Pearls	-2	280	0.7%	1.1%	0%
Pearls	-1	295	1.0%	1.2%	67%
Pearls	0	307	1.3%	1.2%	25%
Basket	-2	72	1.4%	0.3%	0%
Basket	-1	93	0.0%	0.3%	-
Basket	0	70	1.4%	0.3%	100%

AMETHYST (position limit 50)

Players: Market-maker bot + pure noise taker. **Implication:** Ideal counterparty. Quote both sides at or just inside the touch; no adverse selection to price in.



Key findings

- Sizes uniform on 1-4 on all days (chi-square p 0.25-0.65); count dispersion 0.90-1.28; sign ACF(1) -0.01 to 0.10: a textbook Poisson noise trader.
- Side ignores the 10,000 anchor (corr -0.01 to -0.05): P(buy) is 0.48 whether the ask is 10,001 or 10,002.
- Aggressor markout is -1.5 at every horizon from 1 to 500 ticks and price impact is ~0 (VAR long-run impact -0.01 to +0.02 half spreads): the liquidity provider keeps the full half spread.

Findings by day

- **Day -2:** 136 prints / 358 lots, buys 44% (p = 0.20); arrival dispersion 0.90, sign ACF(1) +0.05; side is state-independent (corr with EWMA deviation -0.02); aggressor markout -1.5 at 1 tick and -1.5 at 100 ticks vs half spread 1.5; dominant cluster 'Sparse background'.
- **Day -1:** 159 prints / 396 lots, buys 48% (p = 0.75); arrival dispersion 1.02, sign ACF(1) -0.00; side is state-independent (corr with EWMA deviation -0.03); aggressor markout -1.5 at 1 tick and -1.5 at 100 ticks vs half spread 1.5; dominant cluster 'Sparse background'.
- **Day 0:** 135 prints / 323 lots, buys 51% (p = 0.86); arrival dispersion 1.28, sign ACF(1) +0.10; side is state-independent (corr with EWMA deviation -0.07); aggressor markout -1.5 at 1 tick and -1.5 at 100 ticks vs half spread 1.5; dominant cluster 'Sparse background'.

Metric	Day -2	Day -1	Day 0
Prints / lots (avg size)	136 / 358 (2.63)	159 / 396 (2.49)	135 / 323 (2.39)
Size range / uniform p	1-4 / 0.249	1-4 / 0.451	1-4 / 0.654
L2+ sweeps / multi-print ts	0% / 0%	0% / 0%	0% / 0%
Lots traded / position limit	7.2x	7.9x	6.5x
Count dispersion / gap CV	0.90 / 1.00	1.02 / 0.99	1.28 / 1.11
Buy share (binomial p)	44% (0.20)	48% (0.75)	51% (0.86)
Sign ACF(1) / runs ratio	+0.05 / 0.94	-0.00 / 0.99	+0.10 / 0.89
corr(side, EWMA dev / mom / range)	-0.02 / -0.08 / -0.01	-0.03 / -0.06 / -0.04	-0.07 / -0.06 / -0.05
Markout h1 / h20 / h100 / h500	-1.5 / -1.4 / -1.5 / -1.5	-1.5 / -1.5 / -1.5 / -1.5	-1.5 / -1.4 / -1.5 / -1.4
Eff. half spread / realized / impact (100)	1.48 / +1.51 / -0.03	1.53 / +1.53 / -0.00	1.47 / +1.46 / +0.01
Hasbrouck LR impact (half spreads)	-0.01	+0.02	-0.02
PIN (EKOP)	0.00	0.48	0.31
Top archetype clusters	Sparse background 49%; Max-size prints 26%	Sparse background 45%; Max-size prints 22%	Sparse background 48%; Max-size prints 23%

PAPER_LANTERNS (position limit 50)

Players: Market-maker bot + pure noise taker. **Implication:** Same as AMETHYST: flow is safe to make against; price noise, not flow, sets the edge.



Key findings

- Sizes uniform 1-4; 101-104 prints/day with very regular arrivals (dispersion 0.97-1.20).
- Side is uncorrelated with EWMA deviation, momentum and range position (all |corr| < 0.13).
- Markout is flat at -1.8 to -2.3 from 1 to 100 ticks; realized spread ~2 equals the effective half spread.

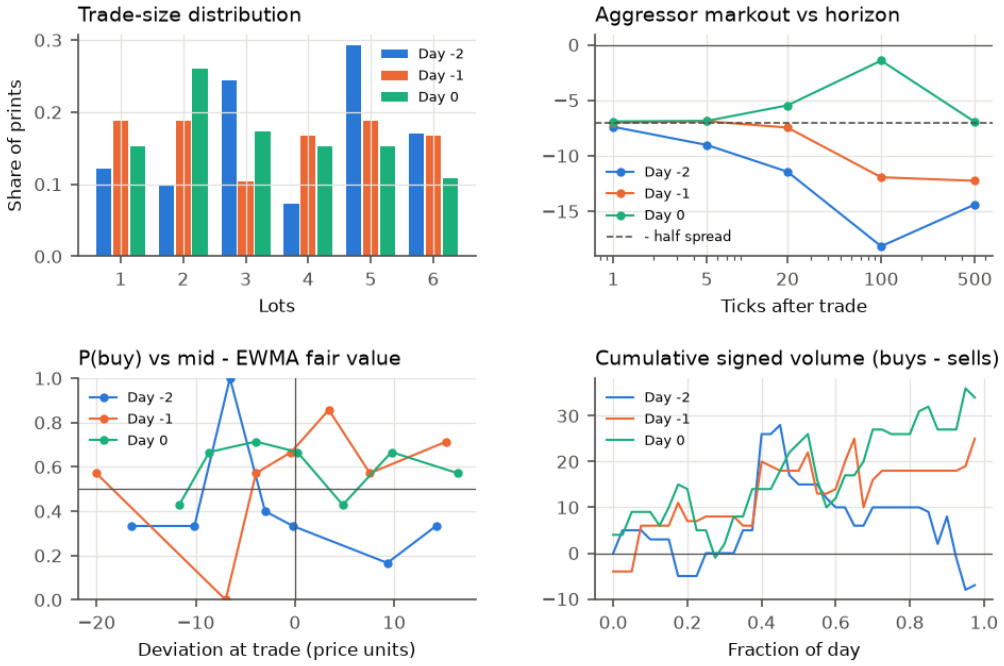
Findings by day

- **Day -2:** 104 prints / 264 lots, buys 41% (p = 0.10); arrival dispersion 1.20, sign ACF(1) -0.05; side is state-independent (corr with EWMA deviation -0.00); aggressor markout -2.0 at 1 tick and -2.1 at 100 ticks vs half spread 2.0; dominant cluster 'Sparse background'.
- **Day -1:** 101 prints / 255 lots, buys 50% (p = 1.00); arrival dispersion 0.99, sign ACF(1) -0.26; side is state-independent (corr with EWMA deviation +0.07); aggressor markout -2.0 at 1 tick and -1.8 at 100 ticks vs half spread 2.0; dominant cluster 'Sparse background'.
- **Day 0:** 103 prints / 252 lots, buys 52% (p = 0.69); arrival dispersion 0.97, sign ACF(1) -0.04; side is state-independent (corr with EWMA deviation -0.08); aggressor markout -1.9 at 1 tick and -2.3 at 100 ticks vs half spread 2.0; dominant cluster 'Sparse background'.

Metric	Day -2	Day -1	Day 0
Prints / lots (avg size)	104 / 264 (2.54)	101 / 255 (2.52)	103 / 252 (2.45)
Size range / uniform p	1-4 / 0.910	1-4 / 0.478	1-4 / 0.848
L2+ sweeps / multi-print ts	0% / 0%	0% / 0%	0% / 0%
Lots traded / position limit	5.3x	5.1x	5.0x
Count dispersion / gap CV	1.20 / 0.97	0.99 / 1.00	0.97 / 0.93
Buy share (binomial p)	41% (0.10)	50% (1.00)	52% (0.69)
Sign ACF(1) / runs ratio	-0.05 / 1.03	-0.26 / 1.24	-0.04 / 1.03
corr(side, EWMA dev / mom / range)	-0.00 / -0.03 / +0.09	+0.07 / +0.12 / +0.08	-0.08 / -0.12 / -0.08
Markout h1 / h20 / h100 / h500	-2.0 / -2.0 / -2.1 / -2.2	-2.0 / -2.1 / -1.8 / +1.2	-1.9 / -2.1 / -2.3 / -2.7
Eff. half spread / realized / impact (100)	2.03 / +2.11 / -0.08	2.04 / +1.84 / +0.20	1.99 / +2.26 / -0.27
Hasbrouck LR impact (half spreads)	+0.11	+0.03	-0.00
PIN (EKOP)	0.00	0.00	0.00
Top archetype clusters	Sparse background 53%; Max-size prints 25%	Sparse background 48%; Max-size prints 22%	Sparse background 57%; Max-size prints 21%

TOFFEE_CAKE (position limit 50)

Players: Market-maker bot + noise taker with clustered arrivals. **Implication:** Making against this flow earns more than the spread on two of three days.



Key findings

- Sizes uniform 1-6 (the widest range among noise products); 41-48 prints/day.
- Arrivals are clustered (dispersion 1.8-2.3, highest outside vouchers/TELLURIUM), but side correlations with state flip sign across days, so there is no stable directional rule.
- Aggressor markout worsens from -7 at 1 tick to -12 to -18 at 100 ticks on days -2/-1: takers lose more than the 7-point half spread as the mean-reverting price moves against them.

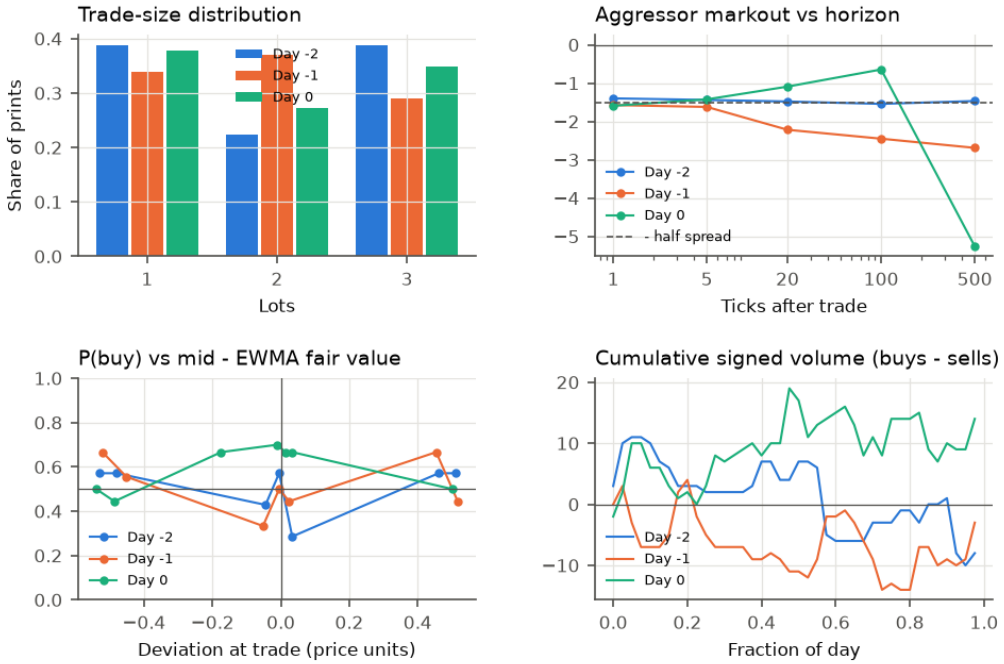
Findings by day

- **Day -2:** 41 prints / 157 lots, buys 41% (p = 0.35); arrival dispersion 1.88, sign ACF(1) +0.12; side is state-independent (corr with EWMA deviation -0.17); aggressor markout -7.4 at 1 tick and -18.2 at 100 ticks vs half spread 7.1; dominant cluster 'Sparse background'.
- **Day -1:** 48 prints / 167 lots, buys 56% (p = 0.47); arrival dispersion 2.34, sign ACF(1) +0.18; side is momentum-leaning (corr with EWMA deviation +0.26); aggressor markout -7.0 at 1 tick and -11.9 at 100 ticks vs half spread 7.2; dominant cluster 'Sparse background'.
- **Day 0:** 46 prints / 148 lots, buys 59% (p = 0.30); arrival dispersion 1.82, sign ACF(1) +0.13; side is state-independent (corr with EWMA deviation -0.08); aggressor markout -6.9 at 1 tick and -1.4 at 100 ticks vs half spread 6.7; dominant cluster 'Sparse background'.

Metric	Day -2	Day -1	Day 0
Prints / lots (avg size)	41 / 157 (3.83)	48 / 167 (3.48)	46 / 148 (3.22)
Size range / uniform p	1-6 / 0.102	1-6 / 0.913	1-6 / 0.614
L2+ sweeps / multi-print ts	0% / 0%	0% / 0%	0% / 0%
Lots traded / position limit	3.1x	3.3x	3.0x
Count dispersion / gap CV	1.88 / 1.39	2.34 / 1.40	1.82 / 1.06
Buy share (binomial p)	41% (0.35)	56% (0.47)	59% (0.30)
Sign ACF(1) / runs ratio	+0.12 / 0.86	+0.18 / 0.81	+0.13 / 0.86
corr(side, EWMA dev / mom / range)	-0.17 / -0.06 / -0.06	+0.26 / +0.16 / +0.20	-0.08 / -0.29 / +0.04
Markout h1 / h20 / h100 / h500	-7.4 / -11.4 / -18.2 / -14.4	-7.0 / -7.5 / -11.9 / -12.3	-6.9 / -5.5 / -1.4 / -6.9
Eff. half spread / realized / impact (100)	7.15 / +18.02 / -10.87	7.16 / +11.94 / -4.79	6.74 / +1.42 / +5.32
Hasbrouck LR impact (half spreads)	-0.34	+0.04	-0.09
PIN (EKOP)	0.35 (low n)	0.52 (low n)	0.31 (low n)
Top archetype clusters	Sparse background 49%; Mixed background 32%	Sparse background 50%; Mixed background 25%	Sparse background 67%; Mixed background 22%

GOLDEN_PEARLS (position limit 20)

Players: Market-maker bot + noise taker. **Implication:** Flow is benign; the risk is the steps themselves, not the counterparties.



Key findings

- Sizes uniform 1-3; 49-66 prints/day; Poisson-like arrivals.
- Takers do not anticipate the day -1/0 step-downs: buy share 45-58%, and takers lose 2.5 at 100 ticks on day -1 vs 1.5 on flat day -2.
- Markout profile on day -2 matches AMETHYST (flat -1.5).

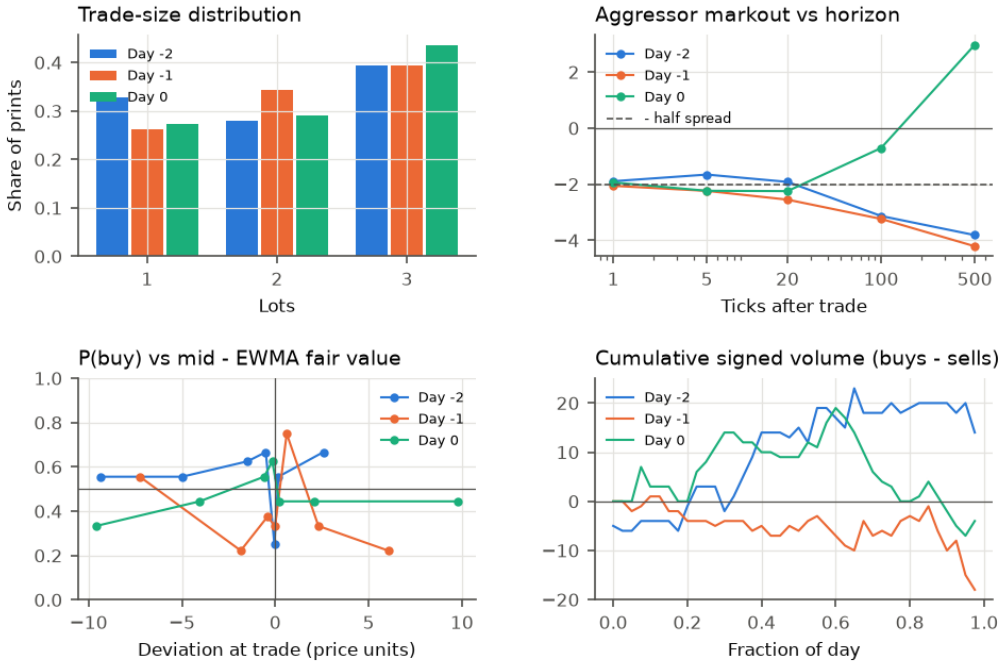
Findings by day

- **Day -2:** 49 prints / 98 lots, buys 51% (p = 1.00); arrival dispersion 1.13, sign ACF(1) +0.00; side is state-independent (corr with EWMA deviation +0.00); aggressor markout -1.4 at 1 tick and -1.5 at 100 ticks vs half spread 1.5; dominant cluster 'Sparse background'.
- **Day -1:** 62 prints / 121 lots, buys 52% (p = 0.90); arrival dispersion 0.87, sign ACF(1) +0.02; side is state-independent (corr with EWMA deviation -0.15); aggressor markout -1.6 at 1 tick and -2.5 at 100 ticks vs half spread 1.5; dominant cluster 'Sparse background'.
- **Day 0:** 66 prints / 130 lots, buys 59% (p = 0.18); arrival dispersion 0.93, sign ACF(1) +0.24; side is state-independent (corr with EWMA deviation +0.13); aggressor markout -1.6 at 1 tick and -0.6 at 100 ticks vs half spread 1.5; dominant cluster 'Sparse background'.

Metric	Day -2	Day -1	Day 0
Prints / lots (avg size)	49 / 98 (2.00)	62 / 121 (1.95)	66 / 130 (1.97)
Size range / uniform p	1-3 / 0.271	1-3 / 0.736	1-3 / 0.554
L2+ sweeps / multi-print ts	0% / 0%	0% / 0%	0% / 0%
Lots traded / position limit	4.9x	6.0x	6.5x
Count dispersion / gap CV	1.13 / 1.07	0.87 / 0.92	0.93 / 0.92
Buy share (binomial p)	51% (1.00)	52% (0.90)	59% (0.18)
Sign ACF(1) / runs ratio	+0.00 / 0.98	+0.02 / 0.97	+0.24 / 0.76
corr(side, EWMA dev / mom / range)	+0.00 / -0.02 / +0.00	-0.15 / -0.11 / -0.07	+0.13 / +0.14 / +0.16
Markout h1 / h20 / h100 / h500	-1.4 / -1.5 / -1.5 / -1.5	-1.6 / -2.2 / -2.5 / -2.7	-1.6 / -1.1 / -0.6 / -5.3
Eff. half spread / realized / impact (100)	1.49 / +1.54 / -0.05	1.54 / +2.45 / -0.91	1.51 / +0.64 / +0.86
Hasbrouck LR impact (half spreads)	-0.02	-0.07	+0.12
PIN (EKOP)	0.24 (low n)	0.66 (low n)	0.06 (low n)
Top archetype clusters	Sparse background 43%; Max-size prints 39%	Sparse background 61%; Max-size prints 27%	Sparse background 44%; Max-size prints 35%

PINK_PEARLS (position limit 20)

Players: Market-maker bot + noise taker. **Implication:** The pearl-sum edge from the EDA report is uncontested by other takers.



Key findings

- Sizes uniform 1-3; 61-62 prints/day; Poisson-like.
- Takers ignore the pearl-sum signal that predicts PINK's jumps: corr(side, 50,000 - pearl sum) is -0.09 to -0.01. Nobody in the tape trades the 50,000 constraint.
- Markout -1.9 to -2.1 at 1 tick, worsening to -3 to -4 at 100-500 ticks on days -2/-1 as jumps land.

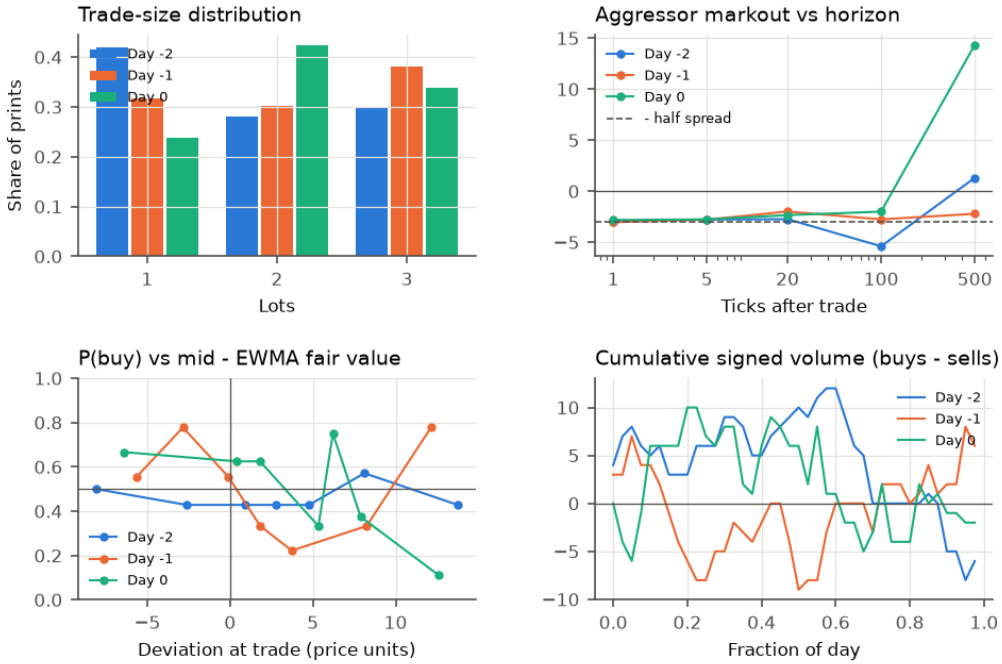
Findings by day

- **Day -2:** 61 prints / 126 lots, buys 56% (p = 0.44); arrival dispersion 1.06, sign ACF(1) +0.09; side is state-independent (corr with EWMA deviation -0.01); aggressor markout -1.9 at 1 tick and -3.1 at 100 ticks vs half spread 2.0; dominant cluster 'Sparse background'.
- **Day -1:** 61 prints / 130 lots, buys 39% (p = 0.12); arrival dispersion 0.96, sign ACF(1) -0.25; side is state-independent (corr with EWMA deviation -0.12); aggressor markout -2.1 at 1 tick and -3.2 at 100 ticks vs half spread 2.1; dominant cluster 'Sparse background'.
- **Day 0:** 62 prints / 134 lots, buys 47% (p = 0.70); arrival dispersion 0.87, sign ACF(1) +0.21; side is state-independent (corr with EWMA deviation +0.07); aggressor markout -1.9 at 1 tick and -0.7 at 100 ticks vs half spread 2.0; dominant cluster 'Max-size prints'.

Metric	Day -2	Day -1	Day 0
Prints / lots (avg size)	61 / 126 (2.07)	61 / 130 (2.13)	62 / 134 (2.16)
Size range / uniform p	1-3 / 0.545	1-3 / 0.448	1-3 / 0.230
L2+ sweeps / multi-print ts	0% / 0%	0% / 0%	0% / 0%
Lots traded / position limit	6.3x	6.5x	6.7x
Count dispersion / gap CV	1.06 / 1.17	0.96 / 0.95	0.87 / 1.02
Buy share (binomial p)	56% (0.44)	39% (0.12)	47% (0.70)
Sign ACF(1) / runs ratio	+0.09 / 0.90	-0.25 / 1.23	+0.21 / 0.78
corr(side, EWMA dev / mom / range)	-0.01 / +0.08 / +0.24	-0.12 / -0.03 / +0.10	+0.07 / +0.05 / +0.18
Markout h1 / h20 / h100 / h500	-1.9 / -1.9 / -3.1 / -3.8	-2.1 / -2.5 / -3.2 / -4.2	-1.9 / -2.2 / -0.7 / +3.0
Eff. half spread / realized / impact (100)	1.95 / +3.13 / -1.18	2.09 / +3.25 / -1.16	1.96 / +0.71 / +1.25
Hasbrouck LR impact (half spreads)	+0.14	-0.09	-0.03
PIN (EKOP)	0.57 (low n)	0.00 (low n)	0.61 (low n)
Top archetype clusters	Sparse background 49%; Max-size prints 39%	Sparse background 43%; Max-size prints 39%	Max-size prints 44%; Sparse background 42%

BLUE_PEARLS (position limit 20)

Players: Market-maker bot + noise taker. **Implication:** Lean passive quotes with the trend; the flow will not compete for it.



Key findings

- Sizes uniform 1-3; 50-63 prints/day.
- Despite a +72/+247/+277 daily uptrend, takers buy only 49-54% of the time: they do not trade the trend.
- Long-horizon markouts follow the trend (day 0 500-tick markout +14): buyers gain, sellers lose.

Findings by day

- **Day -2:** 50 prints / 94 lots, buys 46% (p = 0.67); arrival dispersion 0.83, sign ACF(1) -0.32; side is state-independent (corr with EWMA deviation -0.08); aggressor markout -2.9 at 1 tick and -5.4 at 100 ticks vs half spread 3.0; dominant cluster 'Sparse background'.
- **Day -1:** 63 prints / 130 lots, buys 51% (p = 1.00); arrival dispersion 0.89, sign ACF(1) +0.19; side is state-independent (corr with EWMA deviation +0.00); aggressor markout -3.0 at 1 tick and -2.8 at 100 ticks vs half spread 3.0; dominant cluster 'Sparse background'.
- **Day 0:** 59 prints / 124 lots, buys 49% (p = 1.00); arrival dispersion 0.96, sign ACF(1) +0.14; side is value-leaning (corr with EWMA deviation -0.32); aggressor markout -2.8 at 1 tick and -2.0 at 100 ticks vs half spread 2.9; dominant cluster 'Sparse background'.

Metric	Day -2	Day -1	Day 0
Prints / lots (avg size)	50 / 94 (1.88)	63 / 130 (2.06)	59 / 124 (2.10)
Size range / uniform p	1-3 / 0.423	1-3 / 0.717	1-3 / 0.214
L2+ sweeps / multi-print ts	0% / 0%	0% / 0%	0% / 0%
Lots traded / position limit	4.7x	6.5x	6.2x
Count dispersion / gap CV	0.83 / 0.98	0.89 / 0.87	0.96 / 1.00
Buy share (binomial p)	46% (0.67)	51% (1.00)	49% (1.00)
Sign ACF(1) / runs ratio	-0.32 / 1.28	+0.19 / 0.80	+0.14 / 0.85
corr(side, EWMA dev / mom / range)	-0.08 / -0.13 / -0.06	+0.00 / +0.15 / +0.19	-0.32 / -0.13 / -0.57
Markout h1 / h20 / h100 / h500	-2.9 / -2.7 / -5.4 / +1.3	-3.0 / -2.0 / -2.8 / -2.2	-2.8 / -2.3 / -2.0 / +14.3
Eff. half spread / realized / impact (100)	3.01 / +5.39 / -2.38	3.03 / +2.76 / +0.27	2.95 / +1.99 / +0.96
Hasbrouck LR impact (half spreads)	+0.14	+0.17	+0.10
PIN (EKOP)	0.00 (low n)	0.55 (low n)	0.62 (low n)
Top archetype clusters	Sparse background 58%; Max-size prints 30%	Sparse background 54%; Max-size prints 38%	Sparse background 46%; Max-size prints 34%

WHITE_PEARLS (position limit 20)

Players: Market-maker bot + noise taker. **Implication:** Benign flow; the 20-lot limit is the binding constraint, not adverse selection.



Key findings

- Sizes uniform 1-3; 61-67 prints/day; buy share 47-56%.
- No stable state dependence (corr with EWMA deviation +0.03, +0.11, -0.09).
- Markout -1.6 to -2.0 at 1 tick; long-horizon markouts follow the day's drift.

Findings by day

- **Day -2:** 67 prints / 138 lots, buys 46% (p = 0.63); arrival dispersion 0.91, sign ACF(1) -0.22; side is state-independent (corr with EWMA deviation +0.03); aggressor markout -1.6 at 1 tick and -4.3 at 100 ticks vs half spread 2.0; dominant cluster 'Sparse background'.
- **Day -1:** 61 prints / 108 lots, buys 62% (p = 0.07); arrival dispersion 0.99, sign ACF(1) -0.22; side is state-independent (corr with EWMA deviation +0.11); aggressor markout -2.0 at 1 tick and -7.4 at 100 ticks vs half spread 2.1; dominant cluster 'Sparse background'.
- **Day 0:** 66 prints / 133 lots, buys 56% (p = 0.39); arrival dispersion 1.05, sign ACF(1) -0.16; side is state-independent (corr with EWMA deviation -0.09); aggressor markout -1.8 at 1 tick and +3.2 at 100 ticks vs half spread 2.0; dominant cluster 'Sparse background'.

Metric	Day -2	Day -1	Day 0
Prints / lots (avg size)	67 / 138 (2.06)	61 / 108 (1.77)	66 / 133 (2.02)
Size range / uniform p	1-3 / 0.788	1-3 / 0.069	1-3 / 0.422
L2+ sweeps / multi-print ts	0% / 0%	0% / 0%	0% / 0%
Lots traded / position limit	6.9x	5.4x	6.7x
Count dispersion / gap CV	0.91 / 0.89	0.99 / 1.03	1.05 / 0.84
Buy share (binomial p)	46% (0.63)	62% (0.07)	56% (0.39)
Sign ACF(1) / runs ratio	-0.22 / 1.19	-0.22 / 1.18	-0.16 / 1.13
corr(side, EWMA dev / mom / range)	+0.03 / -0.04 / +0.05	+0.11 / +0.06 / +0.09	-0.09 / -0.05 / +0.03
Markout h1 / h20 / h100 / h500	-1.6 / -2.0 / -4.3 / -2.1	-2.0 / -2.9 / -7.4 / -4.8	-1.8 / -2.8 / +3.2 / +3.7
Eff. half spread / realized / impact (100)	1.99 / +4.28 / -2.29	2.07 / +7.41 / -5.34	1.98 / -3.19 / +5.18
Hasbrouck LR impact (half spreads)	+0.14	+0.65	-0.10
PIN (EKOP)	0.00 (low n)	0.06 (low n)	0.06 (low n)
Top archetype clusters	Sparse background 46%; Max-size prints 37%	Sparse background 64%; Max-size prints 20%	Sparse background 58%; Max-size prints 30%

BLACK_PEARLS (position limit 20)

Players: Market-maker bot + noise taker. **Implication:** Among pearls, the best product to provide liquidity in.



Key findings

- Sizes uniform 1-3; 51-58 prints/day.
- Takers lose more than the 4-point half spread at 100 ticks (-5.7 to -6.3): the passive side's realized spread (~6) exceeds the quoted half spread on all three days.
- Side-state correlations are weak and inconsistent (-0.25 to +0.14).

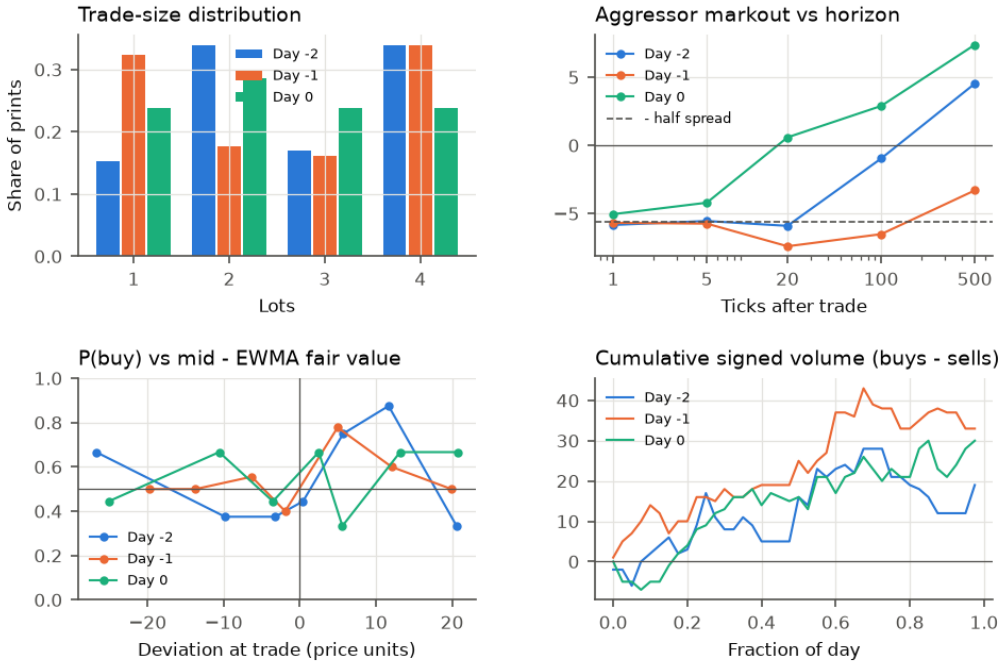
Findings by day

- **Day -2:** 55 prints / 105 lots, buys 38% (p = 0.10); arrival dispersion 1.08, sign ACF(1) -0.01; side is state-independent (corr with EWMA deviation -0.17); aggressor markout -3.6 at 1 tick and -5.9 at 100 ticks vs half spread 3.9; dominant cluster 'Sparse background'.
- **Day -1:** 51 prints / 91 lots, buys 61% (p = 0.16); arrival dispersion 1.13, sign ACF(1) +0.03; side is state-independent (corr with EWMA deviation +0.03); aggressor markout -4.1 at 1 tick and -5.7 at 100 ticks vs half spread 4.0; dominant cluster 'Sparse background'.
- **Day 0:** 58 prints / 112 lots, buys 45% (p = 0.51); arrival dispersion 0.77, sign ACF(1) +0.18; side is value-leaning (corr with EWMA deviation -0.25); aggressor markout -4.1 at 1 tick and -6.3 at 100 ticks vs half spread 4.0; dominant cluster 'Sparse background'.

Metric	Day -2	Day -1	Day 0
Prints / lots (avg size)	55 / 105 (1.91)	51 / 91 (1.78)	58 / 112 (1.93)
Size range / uniform p	1-3 / 0.410	1-3 / 0.113	1-3 / 0.799
L2+ sweeps / multi-print ts	0% / 0%	0% / 0%	0% / 0%
Lots traded / position limit	5.2x	4.5x	5.6x
Count dispersion / gap CV	1.08 / 1.07	1.13 / 0.97	0.77 / 0.81
Buy share (binomial p)	38% (0.10)	61% (0.16)	45% (0.51)
Sign ACF(1) / runs ratio	-0.01 / 1.00	+0.03 / 0.95	+0.18 / 0.81
corr(side, EWMA dev / mom / range)	-0.17 / -0.16 / -0.17	+0.03 / +0.14 / -0.01	-0.25 / -0.28 / -0.15
Markout h1 / h20 / h100 / h500	-3.6 / -6.6 / -5.9 / +1.9	-4.1 / -4.6 / -5.7 / +2.7	-4.1 / -5.7 / -6.3 / -6.9
Eff. half spread / realized / impact (100)	3.93 / +5.94 / -2.01	4.00 / +5.75 / -1.75	3.97 / +6.32 / -2.36
Hasbrouck LR impact (half spreads)	+0.10	-0.10	+0.23
PIN (EKOP)	0.38 (low n)	0.14 (low n)	0.77 (low n)
Top archetype clusters	Sparse background 56%; Max-size prints 25%	Sparse background 63%; Max-size prints 25%	Sparse background 59%; Max-size prints 29%

JELLY (position limit 50)

Players: Market-maker bot + noise taker. **Implication:** Spread capture is real; manage inventory, not counterparty risk.



Key findings

- Sizes uniform 1-4; 59-68 prints/day; Poisson-like.
- The ~5.5 half spread is paid in full at 1 tick (markout -5.0 to -5.8).
- Longer markouts are dominated by random-walk drift (-6.5 to +2.9 at 100 ticks), not flow information.

Findings by day

- **Day -2:** 59 prints / 159 lots, buys 54% (p = 0.60); arrival dispersion 0.96, sign ACF(1) +0.13; side is state-independent (corr with EWMA deviation -0.01); aggressor markout -5.8 at 1 tick and -1.0 at 100 ticks vs half spread 5.6; dominant cluster 'Sparse background'.
- **Day -1:** 68 prints / 171 lots, buys 54% (p = 0.54); arrival dispersion 1.10, sign ACF(1) -0.05; side is state-independent (corr with EWMA deviation +0.08); aggressor markout -5.7 at 1 tick and -6.5 at 100 ticks vs half spread 5.5; dominant cluster 'Sparse background'.
- **Day 0:** 63 prints / 156 lots, buys 56% (p = 0.45); arrival dispersion 0.98, sign ACF(1) -0.01; side is state-independent (corr with EWMA deviation +0.12); aggressor markout -5.0 at 1 tick and +2.9 at 100 ticks vs half spread 5.6; dominant cluster 'Sparse background'.

Metric	Day -2	Day -1	Day 0
Prints / lots (avg size)	59 / 159 (2.69)	68 / 171 (2.51)	63 / 156 (2.48)
Size range / uniform p	1-4 / 0.057	1-4 / 0.066	1-4 / 0.934
L2+ sweeps / multi-print ts	0% / 0%	0% / 0%	0% / 0%
Lots traded / position limit	3.2x	3.4x	3.1x
Count dispersion / gap CV	0.96 / 1.11	1.10 / 0.96	0.98 / 0.81
Buy share (binomial p)	54% (0.60)	54% (0.54)	56% (0.45)
Sign ACF(1) / runs ratio	+0.13 / 0.86	-0.05 / 1.04	-0.01 / 1.00
corr(side, EWMA dev / mom / range)	-0.01 / +0.06 / -0.36	+0.08 / -0.03 / +0.04	+0.12 / -0.17 / +0.02
Markout h1 / h20 / h100 / h500	-5.8 / -5.9 / -1.0 / +4.5	-5.7 / -7.4 / -6.5 / -3.3	-5.0 / +0.6 / +2.9 / +7.4
Eff. half spread / realized / impact (100)	5.64 / +0.97 / +4.68	5.52 / +6.51 / -0.99	5.58 / -2.89 / +8.47
Hasbrouck LR impact (half spreads)	+0.11	+0.03	+0.17
PIN (EKOP)	0.54 (low n)	0.00 (low n)	0.00 (low n)
Top archetype clusters	Sparse background 54%; Max-size prints 34%	Sparse background 51%; Max-size prints 34%	Sparse background 67%; Max-size prints 24%

PEANUT_BUTTER (position limit 50)

Players: Market-maker bot + noise taker. **Implication:** Benign flow in a volatile product; inventory risk dominates.



Key findings

- Sizes uniform 1-4; 58-79 prints/day; day -1 has the only notable buy imbalance (67%, p = 0.24).
- Markout -2.5 to -3.9 at 1 tick; 500-tick markouts swing from -16 to +3 with the trend.
- Hasbrouck long-run impact -0.43 to +0.23 half spreads: no consistent information content.

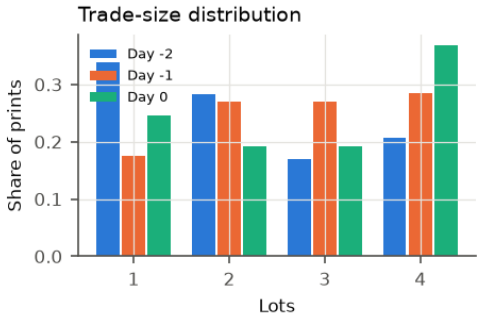
Findings by day

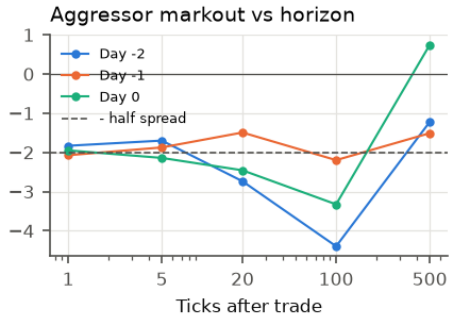
- **Day -2:** 79 prints / 194 lots, buys 53% (p = 0.65); arrival dispersion 0.90, sign ACF(1) +0.07; side is state-independent (corr with EWMA deviation +0.04); aggressor markout -2.5 at 1 tick and +2.8 at 100 ticks vs half spread 3.0; dominant cluster 'Sparse background'.
- **Day -1:** 58 prints / 153 lots, buys 59% (p = 0.24); arrival dispersion 1.68, sign ACF(1) +0.05; side is state-independent (corr with EWMA deviation -0.01); aggressor markout -2.7 at 1 tick and -4.0 at 100 ticks vs half spread 2.9; dominant cluster 'Sparse background'.
- **Day 0:** 59 prints / 148 lots, buys 59% (p = 0.19); arrival dispersion 1.24, sign ACF(1) +0.06; side is state-independent (corr with EWMA deviation -0.19); aggressor markout -3.9 at 1 tick and +0.0 at 100 ticks vs half spread 3.0; dominant cluster 'Sparse background'.

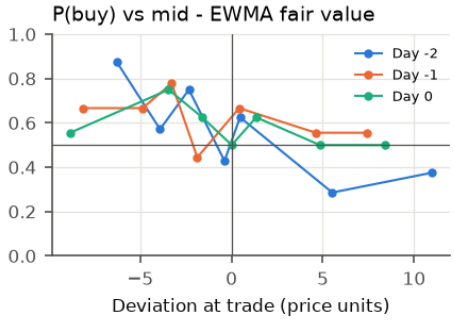
Metric	Day -2	Day -1	Day 0
Prints / lots (avg size)	79 / 194 (2.46)	58 / 153 (2.64)	59 / 148 (2.51)
Size range / uniform p	1-4 / 0.500	1-4 / 0.214	1-4 / 0.980
L2+ sweeps / multi-print ts	0% / 0%	0% / 0%	0% / 0%
Lots traded / position limit	3.9x	3.1x	3.0x
Count dispersion / gap CV	0.90 / 0.95	1.68 / 1.35	1.24 / 1.10
Buy share (binomial p)	53% (0.65)	59% (0.24)	59% (0.19)
Sign ACF(1) / runs ratio	+0.07 / 0.92	+0.05 / 0.93	+0.06 / 0.92
corr(side, EWMA dev / mom / range)	+0.04 / +0.06 / -0.02	-0.01 / -0.01 / -0.01	-0.19 / -0.14 / -0.12
Markout h1 / h20 / h100 / h500	-2.5 / -1.2 / +2.8 / -9.5	-2.7 / -2.6 / -4.0 / -16.1	-3.9 / -0.9 / +0.0 / +3.1
Eff. half spread / realized / impact (100)	3.02 / -2.84 / +5.86	2.91 / +3.98 / -1.07	3.04 / +0.00 / +3.04
Hasbrouck LR impact (half spreads)	+0.23	-0.24	-0.43
PIN (EKOP)	0.00 (low n)	0.14 (low n)	0.03 (low n)
Top archetype clusters	Sparse background 57%; Max-size prints 23%	Sparse background 45%; Max-size prints 34%	Sparse background 66%; Max-size prints 24%

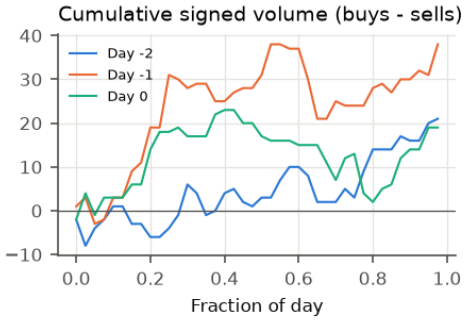
LYCHEE (position limit 200)

Players: Market-maker bot + noise taker. **Implication:** Safe to make and to use as the voucher hedge; voucher takers do not trade LYCHEE momentum.









Key findings

- Sizes uniform 1-4; 53-63 prints/day; the deepest book (L1 10-40).
- Day -2 shows a contrarian lean (corr with EWMA deviation -0.34) that does not repeat on days -1/0.
- Markout ~-2 at 1 tick; price impact small and of unstable sign.

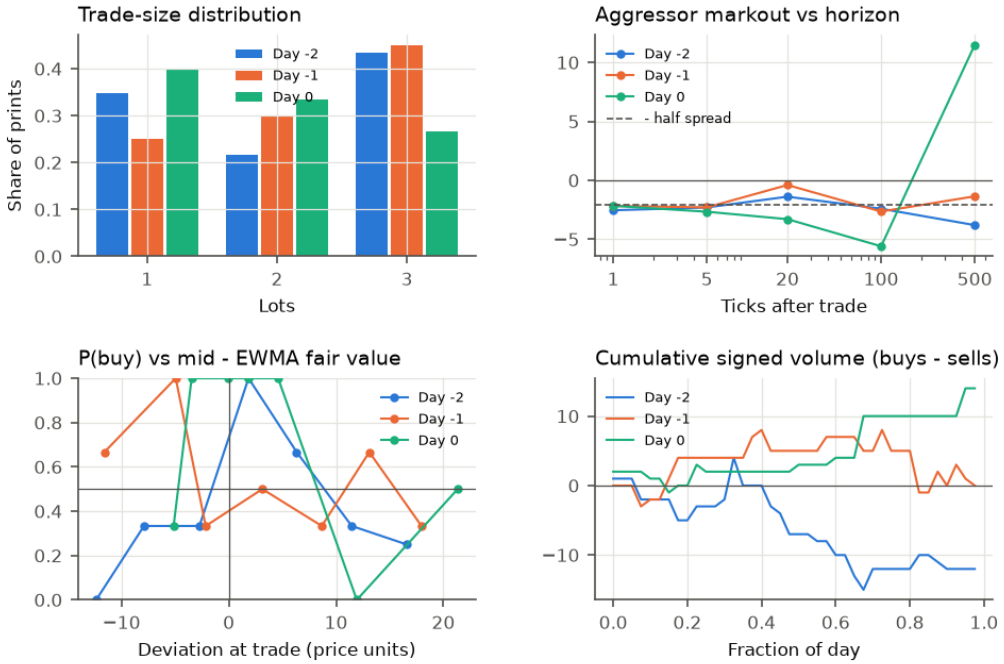
Findings by day

- **Day -2:** 53 prints / 119 lots, buys 57% (p = 0.41); arrival dispersion 0.78, sign ACF(1) +0.18; side is value-leaning (corr with EWMA deviation -0.34); aggressor markout -1.8 at 1 tick and -4.4 at 100 ticks vs half spread 2.0; dominant cluster 'Sparse background'.
- **Day -1:** 63 prints / 168 lots, buys 62% (p = 0.08); arrival dispersion 0.95, sign ACF(1) +0.25; side is state-independent (corr with EWMA deviation -0.13); aggressor markout -2.1 at 1 tick and -2.2 at 100 ticks vs half spread 2.0; dominant cluster 'Sparse background'.
- **Day 0:** 57 prints / 153 lots, buys 58% (p = 0.29); arrival dispersion 1.14, sign ACF(1) -0.31; side is state-independent (corr with EWMA deviation -0.01); aggressor markout -1.9 at 1 tick and -3.3 at 100 ticks vs half spread 2.0; dominant cluster 'Sparse background'.

Metric	Day -2	Day -1	Day 0
Prints / lots (avg size)	53 / 119 (2.25)	63 / 168 (2.67)	57 / 153 (2.68)
Size range / uniform p	1-4 / 0.298	1-4 / 0.582	1-4 / 0.196
L2+ sweeps / multi-print ts	0% / 0%	0% / 0%	0% / 0%
Lots traded / position limit	0.6x	0.8x	0.8x
Count dispersion / gap CV	0.78 / 0.82	0.95 / 0.95	1.14 / 1.12
Buy share (binomial p)	57% (0.41)	62% (0.08)	58% (0.29)
Sign ACF(1) / runs ratio	+0.18 / 0.81	+0.25 / 0.75	-0.31 / 1.29
corr(side, EWMA dev / mom / range)	-0.34 / -0.17 / -0.20	-0.13 / +0.05 / -0.12	-0.01 / -0.03 / +0.02
Markout h1 / h20 / h100 / h500	-1.8 / -2.7 / -4.4 / -1.2	-2.1 / -1.5 / -2.2 / -1.5	-1.9 / -2.5 / -3.3 / +0.7
Eff. half spread / realized / impact (100)	1.99 / +4.39 / -2.40	2.05 / +2.20 / -0.15	1.98 / +3.32 / -1.34
Hasbrouck LR impact (half spreads)	+0.16	-0.16	-0.05
PIN (EKOP)	0.69 (low n)	0.24 (low n)	0.00 (low n)
Top archetype clusters	Sparse background 70%; Max-size prints 21%	Sparse background 60%; Max-size prints 29%	Sparse background 49%; Max-size prints 37%

DRAGONFRUIT (position limit 40)

Players: Market-maker bot + noise taker (thin). **Implication:** Too little flow to identify a player beyond noise; treat as benign.



Key findings

- Only 15-23 prints/day; sizes uniform 1-3; Poisson-like.
- Side-state correlations change sign across days (EWMA deviation +0.22, -0.30, -0.20): consistent with noise at this sample size.
- No co-timing with FRUIT_BASKET beyond chance.

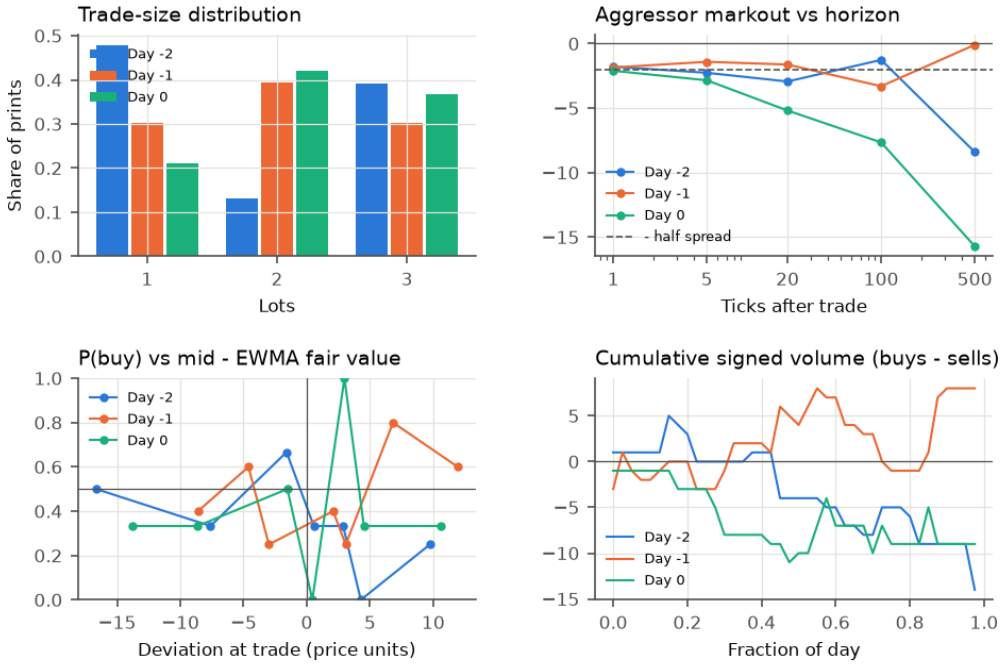
Findings by day

- **Day -2:** 23 prints / 48 lots, buys 39% (p = 0.40); arrival dispersion 0.95, sign ACF(1) -; side is momentum-leaning (corr with EWMA deviation +0.22); aggressor markout -2.5 at 1 tick and -2.4 at 100 ticks vs half spread 2.1; dominant cluster 'Sparse background'.
- **Day -1:** 20 prints / 44 lots, buys 55% (p = 0.82); arrival dispersion 1.01, sign ACF(1) -; side is value-leaning (corr with EWMA deviation -0.30); aggressor markout -2.2 at 1 tick and -2.6 at 100 ticks vs half spread 2.0; dominant cluster 'Sparse background'.
- **Day 0:** 15 prints / 28 lots, buys 67% (p = 0.30); arrival dispersion 0.86, sign ACF(1) -; side is state-independent (corr with EWMA deviation -0.20); aggressor markout -2.2 at 1 tick and -5.6 at 100 ticks vs half spread 2.1; dominant cluster 'Sparse background'.

Metric	Day -2	Day -1	Day 0
Prints / lots (avg size)	23 / 48 (2.09)	20 / 44 (2.20)	15 / 28 (1.87)
Size range / uniform p	1-3 / 0.438	1-3 / 0.522	1-3 / 0.819
L2+ sweeps / multi-print ts	0% / 0%	0% / 0%	0% / 0%
Lots traded / position limit	1.2x	1.1x	0.7x
Count dispersion / gap CV	0.95 / 0.80	1.01 / 1.12	0.86 / 1.09
Buy share (binomial p)	39% (0.40)	55% (0.82)	67% (0.30)
Sign ACF(1) / runs ratio	- / 0.84	- / 1.01	- / 0.91
corr(side, EWMA dev / mom / range)	+0.22 / +0.09 / +0.05	-0.30 / -0.08 / -0.44	-0.20 / -0.36 / -0.12
Markout h1 / h20 / h100 / h500	-2.5 / -1.4 / -2.4 / -3.8	-2.2 / -0.4 / -2.6 / -1.4	-2.2 / -3.3 / -5.6 / +11.5
Eff. half spread / realized / impact (100)	2.09 / +2.41 / -0.33	2.02 / +2.65 / -0.62	2.07 / +5.60 / -3.53
Hasbrouck LR impact (half spreads)	+0.31	-0.14	-0.27
PIN (EKOP)	0.00 (low n)	0.45 (low n)	0.00 (low n)
Top archetype clusters	Sparse background 52%; Max-size prints 43%	Sparse background 55%; Max-size prints 45%	Sparse background 67%; Max-size prints 27%

GRANADILLA (position limit 40)

Players: Market-maker bot + noise taker (thin). **Implication:** Benign but thin.



Key findings

- 19-33 prints/day; sizes uniform 1-3.
- Side-state correlations are inconsistent across days; markout -1.8 to -2.1 at 1 tick.
- No basket-arbitrage footprint.

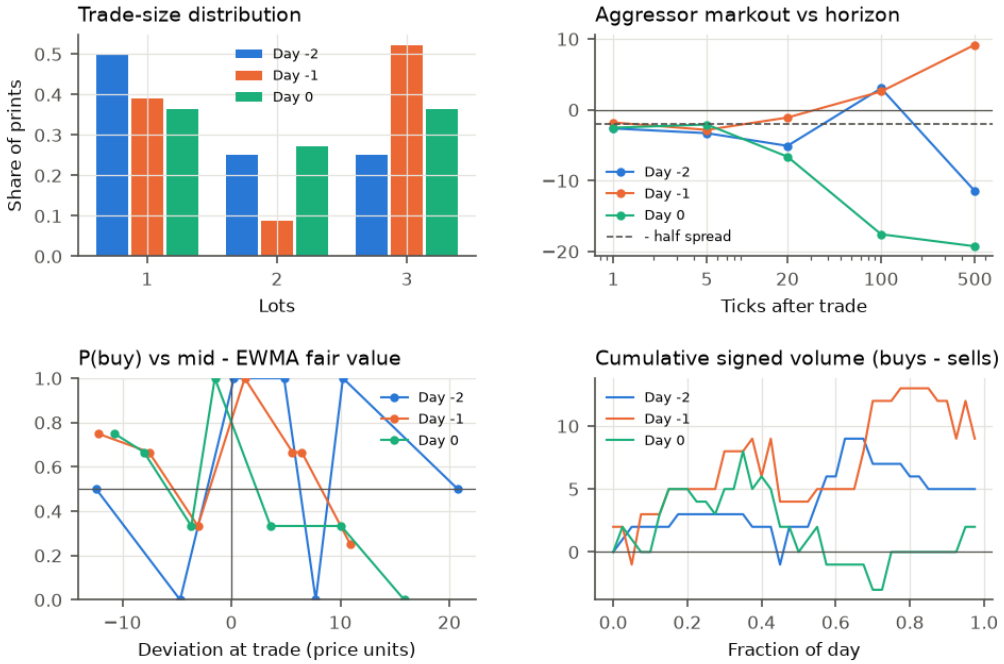
Findings by day

- **Day -2:** 23 prints / 44 lots, buys 35% (p = 0.21); arrival dispersion 1.13, sign ACF(1) -; side is state-independent (corr with EWMA deviation -0.18); aggressor markout -1.8 at 1 tick and -1.3 at 100 ticks vs half spread 2.0; dominant cluster 'Sparse background'.
- **Day -1:** 33 prints / 66 lots, buys 48% (p = 1.00); arrival dispersion 0.86, sign ACF(1) -0.19; side is momentum-leaning (corr with EWMA deviation +0.22); aggressor markout -1.8 at 1 tick and -3.3 at 100 ticks vs half spread 2.0; dominant cluster 'Sparse background'.
- **Day 0:** 19 prints / 41 lots, buys 37% (p = 0.36); arrival dispersion 0.92, sign ACF(1) -; side is state-independent (corr with EWMA deviation +0.08); aggressor markout -2.1 at 1 tick and -7.7 at 100 ticks vs half spread 2.1; dominant cluster 'Sparse background'.

Metric	Day -2	Day -1	Day 0
Prints / lots (avg size)	23 / 44 (1.91)	33 / 66 (2.00)	19 / 41 (2.16)
Size range / uniform p	1-3 / 0.104	1-3 / 0.761	1-3 / 0.504
L2+ sweeps / multi-print ts	0% / 0%	0% / 0%	0% / 0%
Lots traded / position limit	1.1x	1.6x	1.0x
Count dispersion / gap CV	1.13 / 1.17	0.86 / 0.90	0.92 / 0.95
Buy share (binomial p)	35% (0.21)	48% (1.00)	37% (0.36)
Sign ACF(1) / runs ratio	- / 1.05	-0.19 / 1.14	- / 0.71
corr(side, EWMA dev / mom / range)	-0.18 / -0.22 / +0.05	+0.22 / +0.05 / +0.11	+0.08 / +0.33 / -0.11
Markout h1 / h20 / h100 / h500	-1.8 / -2.9 / -1.3 / -8.4	-1.8 / -1.6 / -3.3 / -0.1	-2.1 / -5.2 / -7.7 / -15.7
Eff. half spread / realized / impact (100)	2.02 / +1.26 / +0.76	1.97 / +3.30 / -1.33	2.05 / +7.66 / -5.61
Hasbrouck LR impact (half spreads)	-0.27	+0.32	-0.52
PIN (EKOP)	0.35 (low n)	0.00 (low n)	0.00 (low n)
Top archetype clusters	Sparse background 48%; Max-size prints 39%	Sparse background 67%; Max-size prints 30%	Sparse background 58%; Max-size prints 37%

MANGOSTEEN (position limit 40)

Players: Market-maker bot + noise taker (thin). **Implication:** Benign but thin; statistics are low-power.



Key findings

- Thinnest tape: 12-23 prints/day; sizes uniform 1-3.
- Takers lost 17.6 at 100 ticks on day 0 but gained ~3 on days -2/-1: the sign is unstable.
- No evidence of basket-arbitrage activity.

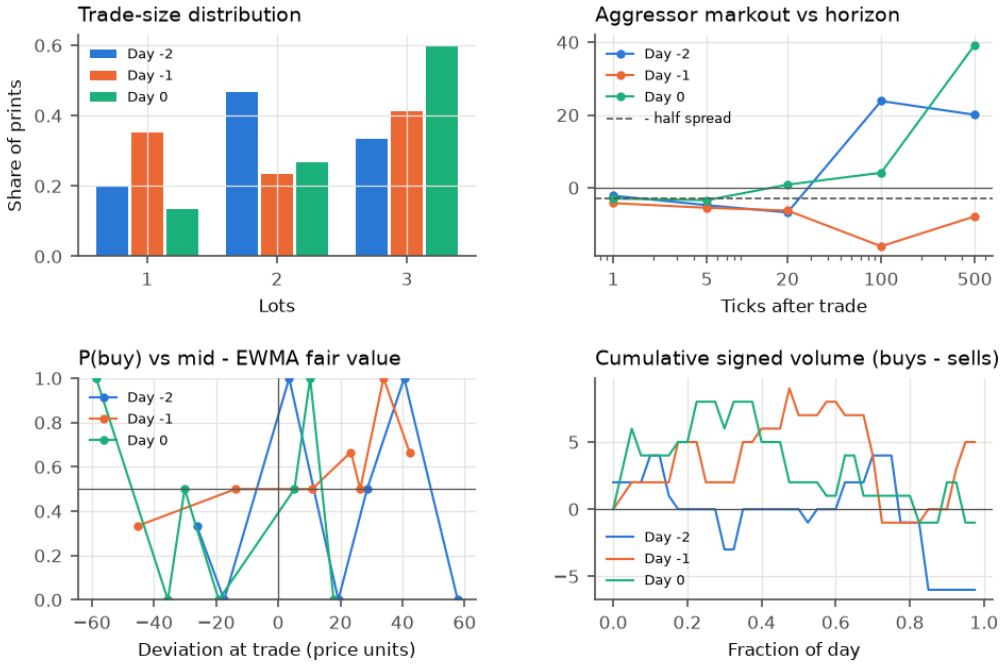
Findings by day

- **Day -2:** 12 prints / 21 lots, buys 58% (p = 0.77); arrival dispersion 0.89, sign ACF(1) -; side is state-independent (corr with EWMA deviation +0.16); aggressor markout -2.6 at 1 tick and +3.0 at 100 ticks vs half spread 2.0; dominant cluster 'Sparse background'.
- **Day -1:** 23 prints / 49 lots, buys 61% (p = 0.40); arrival dispersion 1.13, sign ACF(1) -; side is value-leaning (corr with EWMA deviation -0.24); aggressor markout -1.8 at 1 tick and +2.6 at 100 ticks vs half spread 2.0; dominant cluster 'Max-size prints'.
- **Day 0:** 22 prints / 44 lots, buys 50% (p = 1.00); arrival dispersion 0.79, sign ACF(1) -; side is value-leaning (corr with EWMA deviation -0.32); aggressor markout -2.5 at 1 tick and -17.6 at 100 ticks vs half spread 2.1; dominant cluster 'Sparse background'.

Metric	Day -2	Day -1	Day 0
Prints / lots (avg size)	12 / 21 (1.75)	23 / 49 (2.13)	22 / 44 (2.00)
Size range / uniform p	1-3 / 0.472	1-3 / 0.032	1-3 / 0.834
L2+ sweeps / multi-print ts	0% / 0%	0% / 0%	0% / 0%
Lots traded / position limit	0.5x	1.2x	1.1x
Count dispersion / gap CV	0.89 / 0.67	1.13 / 0.96	0.79 / 0.96
Buy share (binomial p)	58% (0.77)	61% (0.40)	50% (1.00)
Sign ACF(1) / runs ratio	- / 0.59	- / 1.17	- / 1.08
corr(side, EWMA dev / mom / range)	+0.16 / +0.34 / -0.32	-0.24 / -0.31 / -0.11	-0.32 / -0.37 / -0.13
Markout h1 / h20 / h100 / h500	-2.6 / -5.1 / +3.0 / -11.5	-1.8 / -1.1 / +2.6 / +9.2	-2.5 / -6.6 / -17.6 / -19.3
Eff. half spread / realized / impact (100)	1.96 / -3.04 / +5.00	2.00 / -2.61 / +4.61	2.11 / +17.57 / -15.45
Hasbrouck LR impact (half spreads)	-0.68	-0.44	-0.20
PIN (EKOP)	0.00 (low n)	0.39 (low n)	0.00 (low n)
Top archetype clusters	Sparse background 75%; Max-size prints 25%	Max-size prints 52%; Sparse background 43%	Sparse background 64%; Max-size prints 36%

FRUIT_BASKET (position limit 40)

Players: Market-maker bot + noise taker (thin). **Implication:** Nobody arbitrages the basket in the tape; a basket strategy competes only with the bot's quotes.



Key findings

- 15-17 prints/day; sizes uniform 1-3.
- Basket prints are not paired with leg prints: at most one co-timed tick per day across the basket family.
- Markouts are large and unstable (+24, -16, +4 at 100 ticks), reflecting the volatile premium, not information.

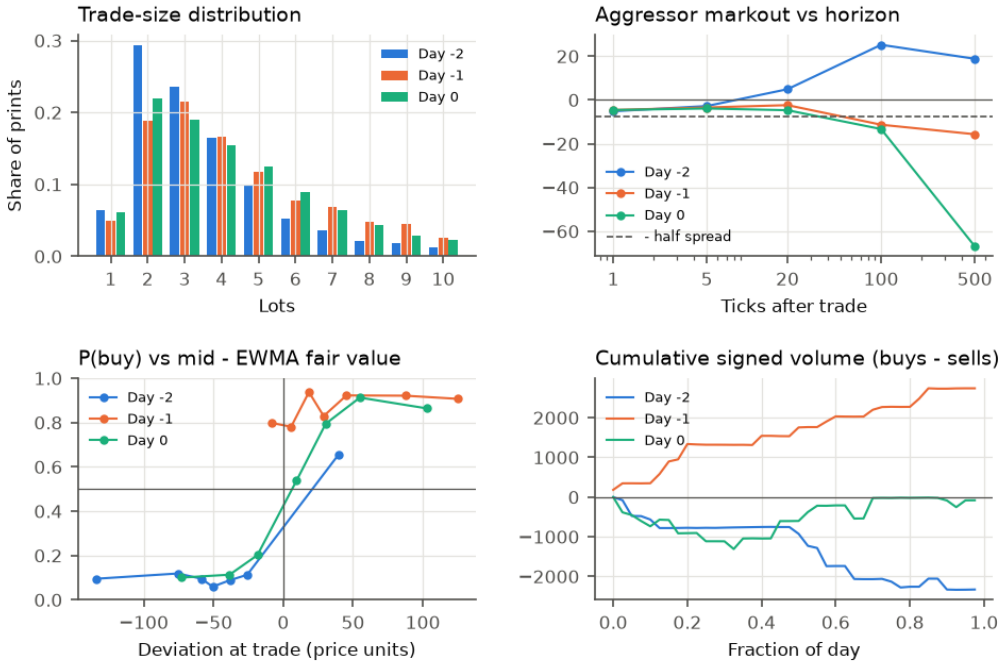
Findings by day

- **Day -2:** 15 prints / 32 lots, buys 40% (p = 0.61); arrival dispersion 0.86, sign ACF(1) -; side is state-independent (corr with EWMA deviation +0.10); aggressor markout -2.2 at 1 tick and +23.8 at 100 ticks vs half spread 3.1; dominant cluster 'Sparse background'.
- **Day -1:** 17 prints / 35 lots, buys 59% (p = 0.63); arrival dispersion 0.96, sign ACF(1) -; side is momentum-leaning (corr with EWMA deviation +0.26); aggressor markout -4.2 at 1 tick and -16.1 at 100 ticks vs half spread 2.9; dominant cluster 'Sparse background'.
- **Day 0:** 15 prints / 37 lots, buys 47% (p = 1.00); arrival dispersion 0.86, sign ACF(1) -; side is value-leaning (corr with EWMA deviation -0.24); aggressor markout -2.9 at 1 tick and +4.1 at 100 ticks vs half spread 2.9; dominant cluster 'Max-size prints'.

Metric	Day -2	Day -1	Day 0
Prints / lots (avg size)	15 / 32 (2.13)	17 / 35 (2.06)	15 / 37 (2.47)
Size range / uniform p	1-3 / 0.449	1-3 / 0.662	1-3 / 0.074
L2+ sweeps / multi-print ts	0% / 0%	0% / 0%	0% / 0%
Lots traded / position limit	0.8x	0.9x	0.9x
Count dispersion / gap CV	0.86 / 0.79	0.96 / 0.63	0.86 / 0.53
Buy share (binomial p)	40% (0.61)	59% (0.63)	47% (1.00)
Sign ACF(1) / runs ratio	- / 0.98	- / 0.97	- / 1.18
corr(side, EWMA dev / mom / range)	+0.10 / +0.19 / -0.37	+0.26 / +0.26 / -0.27	-0.24 / -0.24 / -0.11
Markout h1 / h20 / h100 / h500	-2.2 / -6.8 / +23.8 / +20.1	-4.2 / -6.2 / -16.1 / -7.8	-2.9 / +0.9 / +4.1 / +39.3
Eff. half spread / realized / impact (100)	3.10 / -23.83 / +26.93	2.94 / +16.06 / -13.12	2.93 / -4.13 / +7.07
Hasbrouck LR impact (half spreads)	-1.47	-0.68	-0.81
PIN (EKOP)	0.00 (low n)	1.00 (low n)	0.00 (low n)
Top archetype clusters	Sparse background 67%; Max-size prints 33%	Sparse background 53%; Max-size prints 41%	Max-size prints 60%; Sparse background 40%

TELLURIUM (position limit 40)

Players: Market-maker bot + jump-reactive momentum takers (fast and sweeping) + background noise. **Implication:** Pull or widen quotes for ~200 ticks after a jump; the chasers' one-sided flow runs over resting orders.



Key findings

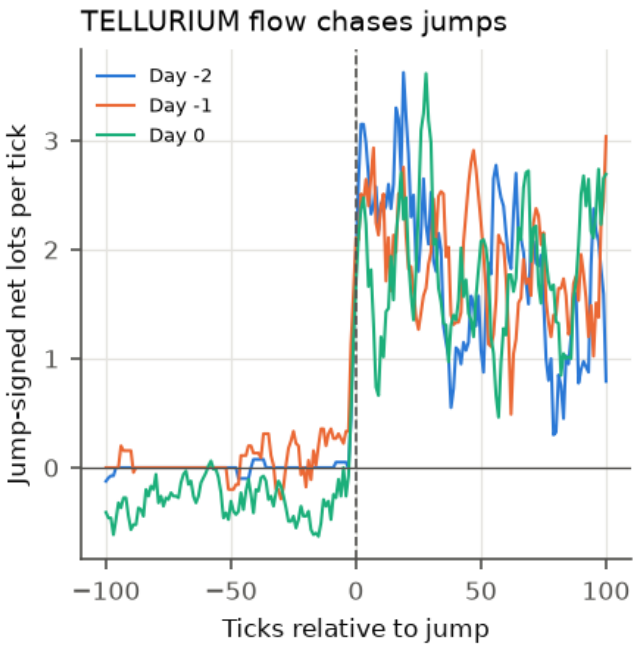
- Non-uniform sizes 1-10 (mode 2-3; $p < 0.001$ vs uniform); 899-1,600 prints/day; the only product with level-2/3 sweeps (23-34% of prints) and multi-print timestamps.
- 87% of prints occur within 200 ticks after a jump, when intensity is 4-6x its quiet level. Jump-signed net flow is ~0 in the 200 ticks before a jump and +287 to +324 lots after: no anticipation, heavy follow-through.
- Side tracks past moves (corr with EWMA deviation +0.10 to +0.58; with range position +0.09 to +0.52) and the next tick's mid (+0.72), not 100-tick future returns. Post-jump prints lose 35 at 500 ticks.
- The mixture model splits the crowd: 'fast momentum' prints (22-39%) earn +15 at 100 ticks; 'sweeping chasers' (23-34%) lose 21 at 500 ticks.

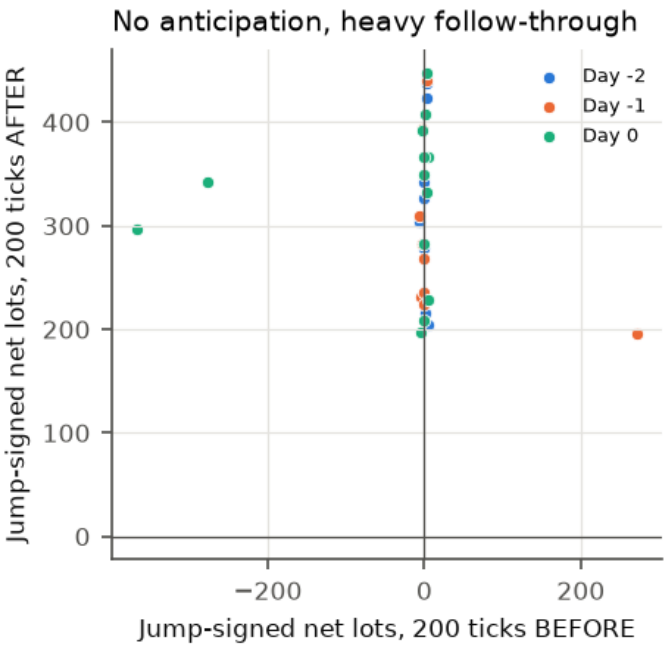
Findings by day

- **Day -2:** 1176 prints / 4180 lots, buys 18% (p = 0.00); arrival dispersion 52.46, sign ACF(1) +0.40; side is momentum-leaning (corr with EWMA deviation +0.44); aggressor markout -5.2 at 1 tick and +25.2 at 100 ticks vs half spread 8.1; dominant cluster 'Fast momentum'.
- **Day -1:** 899 prints / 3875 lots, buys 87% (p = 0.00); arrival dispersion 37.19, sign ACF(1) +0.18; side is state-independent (corr with EWMA deviation +0.10); aggressor markout -4.6 at 1 tick and -11.3 at 100 ticks vs half spread 7.0; dominant cluster 'Fast momentum'.
- **Day 0:** 1600 prints / 6631 lots, buys 50% (p = 0.78); arrival dispersion 39.25, sign ACF(1) +0.62; side is momentum-leaning (corr with EWMA deviation +0.58); aggressor markout -4.6 at 1 tick and -13.3 at 100 ticks vs half spread 7.3; dominant cluster 'Mixed background'.

Metric	Day -2	Day -1	Day 0
Prints / lots (avg size)	1176 / 4180 (3.55)	899 / 3875 (4.31)	1600 / 6631 (4.14)
Size range / uniform p	1-10 / 0.000	1-10 / 0.000	1-10 / 0.000
L2+ sweeps / multi-print ts	34% / 62%	23% / 43%	27% / 51%
Lots traded / position limit	104.5x	96.9x	165.8x
Count dispersion / gap CV	52.46 / 3.80	37.19 / 3.38	39.25 / 3.62
Buy share (binomial p)	18% (0.00)	87% (0.00)	50% (0.78)
Sign ACF(1) / runs ratio	+0.40 / 0.60	+0.18 / 0.82	+0.62 / 0.38
corr(side, EWMA dev / mom / range)	+0.44 / +0.26 / +0.51	+0.10 / +0.04 / +0.09	+0.58 / +0.27 / +0.52
Markout h1 / h20 / h100 / h500	-5.2 / +4.9 / +25.2 / +18.8	-4.6 / -2.4 / -11.3 / -15.7	-4.6 / -4.7 / -13.3 / -66.8
Eff. half spread / realized / impact (100)	8.08 / -25.21 / +33.29	7.00 / +11.31 / -4.30	7.30 / +13.29 / -5.99
Hasbrouck LR impact (half spreads)	+1.46	+1.08	+1.02
PIN (EKOP)	0.83	0.80	0.74
Top archetype clusters	Fast momentum 39%; Sweeping chaser 34%	Fast momentum 31%; Mixed background 29%	Mixed background 33%; Sweeping chaser 27%

TELLURIUM jump event study

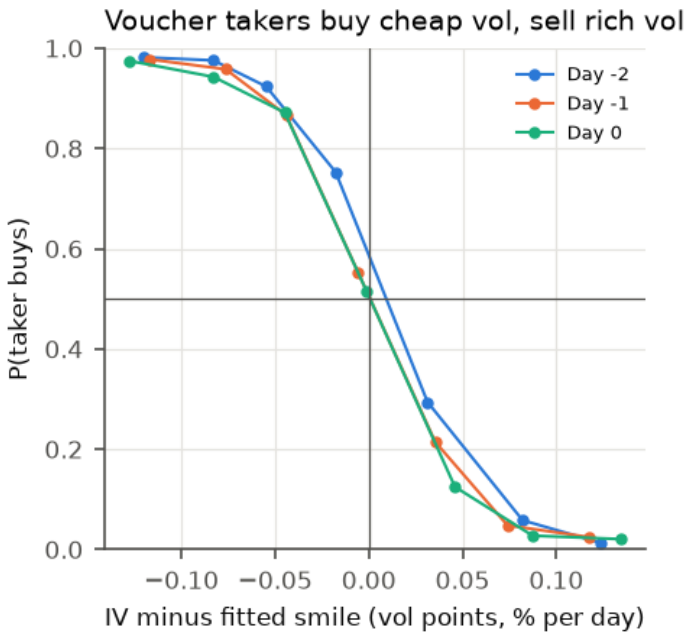


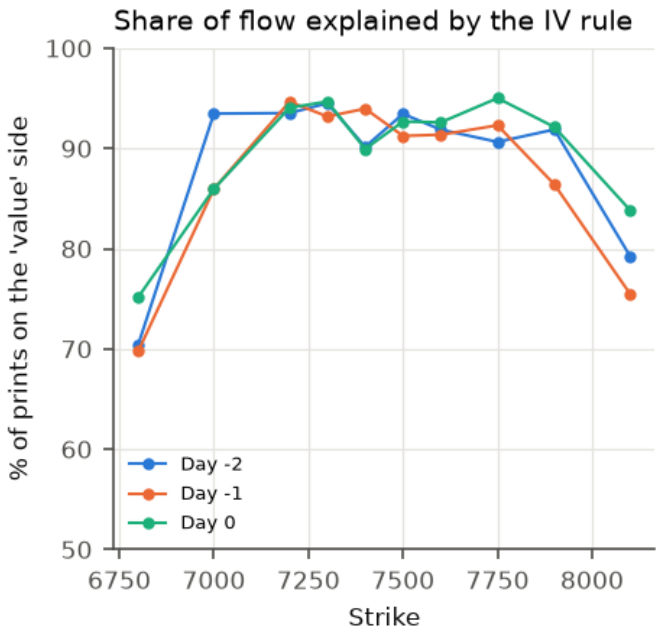


Average per jump (200-tick windows)

Day	Jumps	Mean |jump|	Net lots before (jump-signed)	Net lots after	Prints before	Prints after
-2	8	156	+1	+316	1	125
-1	9	172	+29	+287	10	82
0	13	178	-49	+324	25	114

Family: LYCHEE vouchers - the IV-smile value taker





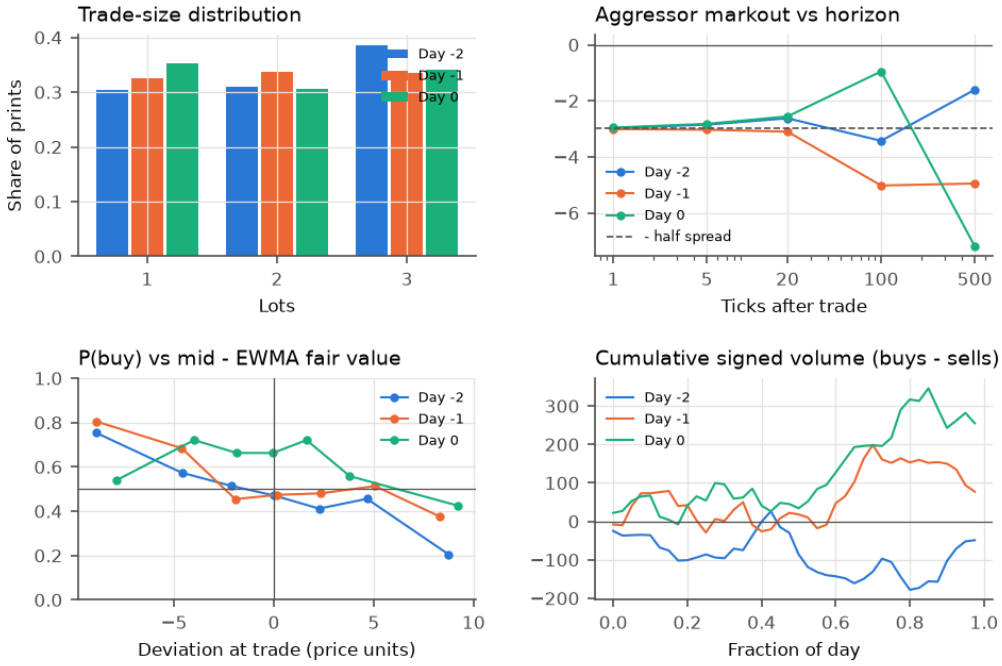
- One rule-based taker dominates all ten strikes: side vs IV-minus-smile residual has corr -0.75 to -0.87 at 7,000-7,900 and -0.47 to -0.71 at the wings (6,800 and 8,100) on every day.
- 86-95% of prints at 7,000-7,900 are on the rule's side, rising to 98% in the top quintile of |residual|. The remaining prints behave like noise (rule-inconsistent prints lose 1.5 at 100 ticks).
- The rule pays where the smile is best identified: rule-consistent prints earn +1.2 to +3.5 per lot at 100 ticks at 7,300-7,600, 0 to +1.2 at 7,200 and 7,900, and lose at 6,800 and 8,100.
- Footprints: long same-side runs (sign ACF(1) 0.67-0.83; runs ratio 0.17-0.33), clustered arrivals (dispersion 2.3-4.5) and PIN 0.57-0.85. These look like metaorders (Lillo & Farmer 2004) but are explained by a persistent signal; here PIN measures one-sided flow, not news (Duarte & Young 2009).
- No cross-strike coordination: co-timed prints occur at the independent-arrival rate (23-26% of voucher trade ticks vs 24-26% expected) with 45-53% same-side. The rule is applied strike by strike.
- Takers do not trade LYCHEE momentum (corr -0.04 to +0.01): they price relative value, not delta.

IV rule by strike and day

Strike	Day	corr(side, IV residual)	Rule-side share	mo100 rule-side	mo100 other
6800	-2	-0.55	70%	-3.28	-3.77
	-1	-0.49	70%	-5.68	-3.47
	0	-0.47	75%	-3.35	+6.32
7000	-2	-0.84	93%	-2.87	-0.76
	-1	-0.76	86%	-1.27	-3.85
	0	-0.79	86%	+4.40	-2.80
7200	-2	-0.86	94%	-0.06	+1.15
	-1	-0.86	95%	+0.07	-1.68
	0	-0.87	94%	+1.19	-4.47
7300	-2	-0.86	95%	+3.17	-3.50
	-1	-0.84	93%	+1.20	-1.35
	0	-0.85	95%	+2.51	-3.44
7400	-2	-0.79	90%	+2.45	-3.90
	-1	-0.82	94%	+3.46	+0.29
	0	-0.78	90%	+1.90	-3.54
7500	-2	-0.84	93%	+3.53	-2.39
	-1	-0.82	91%	+2.32	-3.24
	0	-0.84	93%	+1.43	-2.93
7600	-2	-0.81	92%	+3.25	-2.50
	-1	-0.80	91%	+1.43	-2.89
	0	-0.81	93%	+1.70	-1.22
7750	-2	-0.76	91%	+0.58	-2.58
	-1	-0.82	92%	+2.24	-1.38
	0	-0.86	95%	+0.56	-2.25
7900	-2	-0.83	92%	+0.84	-0.70
	-1	-0.75	86%	+0.60	-1.21
	0	-0.81	92%	+0.51	-0.86
8100	-2	-0.58	79%	-0.32	-0.60
	-1	-0.59	76%	-0.41	-0.74
	0	-0.71	84%	-0.56	-0.62

VOUCHER_6800 (position limit 50)

Players: Market-maker bot + IV-smile value taker (most prints) + residual noise taker. **Implication:** Avoid being the passive side to rule-consistent flow; skew quotes by the IV residual (see the vouchers family page)



Findings by day

- **Day -2:** 477 prints / 993 lots, buys 48% (p = 0.52); arrival dispersion 4.04, sign ACF(1) +0.75; side is value-leaning (corr with EWMA deviation -0.31); aggressor markout -3.0 at 1 tick and -3.4 at 100 ticks vs half spread 3.0; dominant cluster 'Max-size prints'.
- **Day -1:** 536 prints / 1077 lots, buys 54% (p = 0.06); arrival dispersion 3.55, sign ACF(1) +0.76; side is value-leaning (corr with EWMA deviation -0.24); aggressor markout -3.0 at 1 tick and -5.0 at 100 ticks vs half spread 3.0; dominant cluster 'Mixed background'.
- **Day 0:** 604 prints / 1201 lots, buys 61% (p = 0.00); arrival dispersion 3.40, sign ACF(1) +0.81; side is state-independent (corr with EWMA deviation -0.06); aggressor markout -3.0 at 1 tick and -1.0 at 100 ticks vs half spread 3.0; dominant cluster 'Mixed background'.

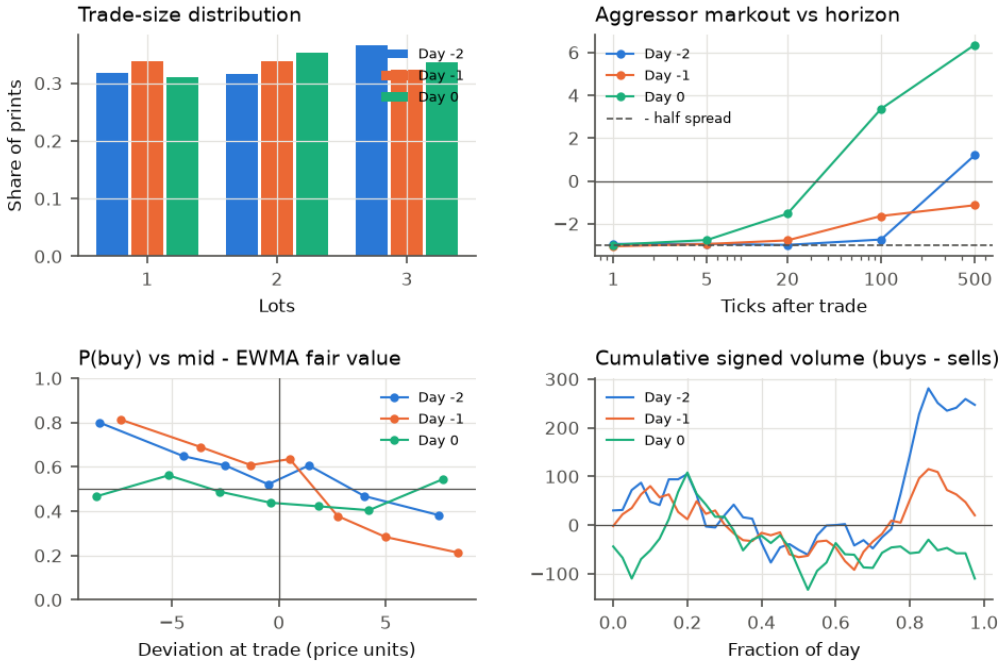
Metric	Day -2	Day -1	Day 0
Prints / lots (avg size)	477 / 993 (2.08)	536 / 1077 (2.01)	604 / 1201 (1.99)
Size range / uniform p	1-3 / 0.052	1-3 / 0.944	1-3 / 0.348
L2+ sweeps / multi-print ts	0% / 0%	0% / 0%	0% / 0%
Lots traded / position limit	19.9x	21.5x	24.0x
Count dispersion / gap CV	4.04 / 1.78	3.55 / 1.79	3.40 / 1.73
Buy share (binomial p)	48% (0.52)	54% (0.06)	61% (0.00)
Sign ACF(1) / runs ratio	+0.75 / 0.25	+0.76 / 0.24	+0.81 / 0.19
corr(side, EWMA dev / mom / range)	-0.31 / -0.18 / -0.23	-0.24 / -0.15 / -0.18	-0.06 / -0.06 / -0.06
Markout h1 / h20 / h100 / h500	-3.0 / -2.6 / -3.4 / -1.6	-3.0 / -3.1 / -5.0 / -5.0	-3.0 / -2.6 / -1.0 / -7.2
Eff. half spread / realized / impact (100)	2.98 / +3.43 / -0.44	2.99 / +5.02 / -2.03	2.99 / +0.95 / +2.04
Hasbrouck LR impact (half spreads)	+0.02	-0.03	+0.05
PIN (EKOP)	0.67	0.74	0.72
Top archetype clusters	Max-size prints 39%; Mixed background 34%	Mixed background 37%; Max-size prints 34%	Mixed background 47%; Max-size prints 34%

Strike-specific findings

- IV rule: corr(side, IV residual) -0.55 to -0.47; 70-75% of prints on the rule's side across days.
- Rule-consistent prints: 100-tick markout -5.68 to -3.28; other prints -3.77 to +6.32.

VOUCHER_7000 (position limit 50)

Players: Market-maker bot + IV-smile value taker (most prints) + residual noise taker. **Implication:** Avoid being the passive side to rule-consistent flow; skew quotes by the IV residual (see the vouchers family page)



Findings by day

- **Day -2:** 659 prints / 1349 lots, buys 58% (p = 0.00); arrival dispersion 3.95, sign ACF(1) +0.79; side is value-leaning (corr with EWMA deviation -0.24); aggressor markout -2.9 at 1 tick and -2.7 at 100 ticks vs half spread 3.0; dominant cluster 'Max-size prints'.
- **Day -1:** 520 prints / 1032 lots, buys 52% (p = 0.46); arrival dispersion 3.28, sign ACF(1) +0.77; side is value-leaning (corr with EWMA deviation -0.41); aggressor markout -3.1 at 1 tick and -1.6 at 100 ticks vs half spread 3.0; dominant cluster 'Mixed background'.
- **Day 0:** 627 prints / 1270 lots, buys 48% (p = 0.23); arrival dispersion 2.30, sign ACF(1) +0.77; side is state-independent (corr with EWMA deviation +0.00); aggressor markout -3.0 at 1 tick and +3.4 at 100 ticks vs half spread 3.0; dominant cluster 'Mixed background'.

Metric	Day -2	Day -1	Day 0
Prints / lots (avg size)	659 / 1349 (2.05)	520 / 1032 (1.98)	627 / 1270 (2.03)
Size range / uniform p	1-3 / 0.210	1-3 / 0.884	1-3 / 0.439
L2+ sweeps / multi-print ts	0% / 0%	0% / 0%	0% / 0%
Lots traded / position limit	27.0x	20.6x	25.4x
Count dispersion / gap CV	3.95 / 1.87	3.28 / 1.75	2.30 / 1.51
Buy share (binomial p)	58% (0.00)	52% (0.46)	48% (0.23)
Sign ACF(1) / runs ratio	+0.79 / 0.21	+0.77 / 0.23	+0.77 / 0.23
corr(side, EWMA dev / mom / range)	-0.24 / -0.14 / -0.35	-0.41 / -0.19 / -0.29	+0.00 / -0.00 / -0.20
Markout h1 / h20 / h100 / h500	-2.9 / -3.0 / -2.7 / +1.2	-3.1 / -2.8 / -1.6 / -1.1	-3.0 / -1.5 / +3.4 / +6.4
Eff. half spread / realized / impact (100)	3.00 / +2.74 / +0.26	3.00 / +1.64 / +1.36	3.00 / -3.37 / +6.37
Hasbrouck LR impact (half spreads)	+0.01	-0.01	+0.07
PIN (EKOP)	0.78	0.79	0.79
Top archetype clusters	Max-size prints 37%; Mixed background 36%	Mixed background 38%; Max-size prints 32%	Mixed background 43%; Max-size prints 34%

Strike-specific findings

- IV rule: corr(side, IV residual) -0.84 to -0.76; 86-93% of prints on the rule's side across days.
- Rule-consistent prints: 100-tick markout -2.87 to +4.40; other prints -3.85 to -0.76.

VOUCHER_7200 (position limit 50)

Players: Market-maker bot + IV-smile value taker (most prints) + residual noise taker. **Implication:** Avoid being the passive side to rule-consistent flow; skew quotes by the IV residual (see the vouchers family page)



Findings by day

- **Day -2:** 664 prints / 1340 lots, buys 52% (p = 0.46); arrival dispersion 4.48, sign ACF(1) +0.81; side is value-leaning (corr with EWMA deviation -0.34); aggressor markout -1.5 at 1 tick and +0.0 at 100 ticks vs half spread 1.5; dominant cluster 'Mixed background'.
- **Day -1:** 637 prints / 1275 lots, buys 57% (p = 0.00); arrival dispersion 3.27, sign ACF(1) +0.83; side is value-leaning (corr with EWMA deviation -0.44); aggressor markout -1.5 at 1 tick and -0.0 at 100 ticks vs half spread 1.5; dominant cluster 'Mixed background'.
- **Day 0:** 589 prints / 1173 lots, buys 42% (p = 0.00); arrival dispersion 3.30, sign ACF(1) +0.83; side is value-leaning (corr with EWMA deviation -0.22); aggressor markout -1.4 at 1 tick and +0.8 at 100 ticks vs half spread 1.5; dominant cluster 'Mixed background'.

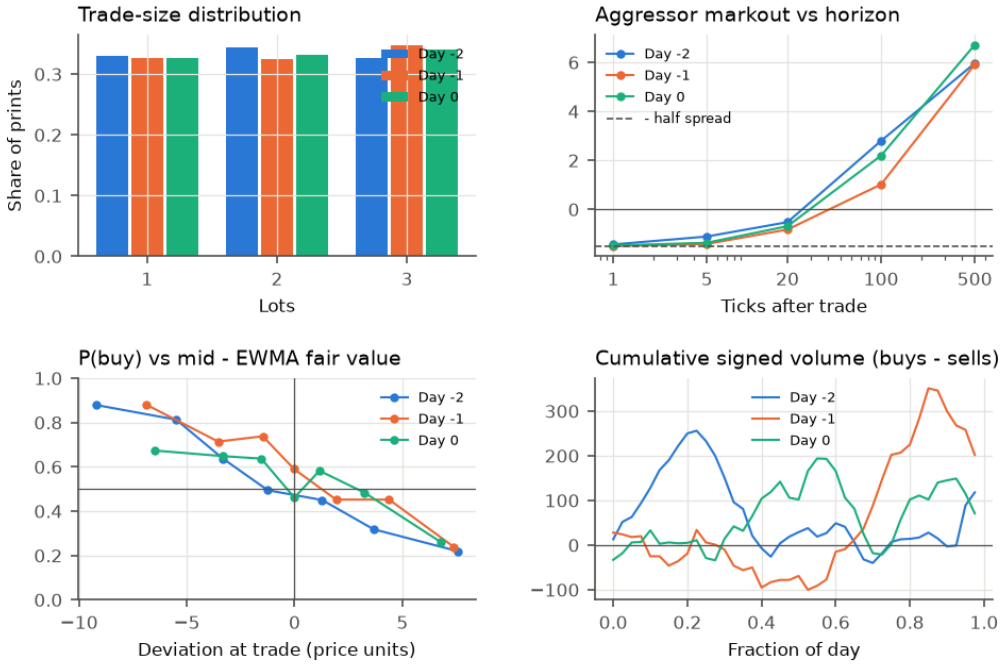
Metric	Day -2	Day -1	Day 0
Prints / lots (avg size)	664 / 1340 (2.02)	637 / 1275 (2.00)	589 / 1173 (1.99)
Size range / uniform p	1-3 / 0.410	1-3 / 0.067	1-3 / 0.968
L2+ sweeps / multi-print ts	0% / 0%	0% / 0%	0% / 0%
Lots traded / position limit	26.8x	25.5x	23.5x
Count dispersion / gap CV	4.48 / 1.93	3.27 / 1.71	3.30 / 1.83
Buy share (binomial p)	52% (0.46)	57% (0.00)	42% (0.00)
Sign ACF(1) / runs ratio	+0.81 / 0.19	+0.83 / 0.17	+0.83 / 0.17
corr(side, EWMA dev / mom / range)	-0.34 / -0.17 / -0.28	-0.44 / -0.22 / -0.30	-0.22 / -0.17 / -0.19
Markout h1 / h20 / h100 / h500	-1.5 / -0.8 / +0.0 / -1.1	-1.5 / -1.0 / -0.0 / +2.3	-1.4 / -0.6 / +0.8 / -1.2
Eff. half spread / realized / impact (100)	1.50 / -0.02 / +1.52	1.50 / +0.02 / +1.48	1.50 / -0.85 / +2.35
Hasbrouck LR impact (half spreads)	+0.15	+0.04	+0.09
PIN (EKOP)	0.77	0.81	0.85
Top archetype clusters	Mixed background 38%; Max-size prints 33%	Mixed background 38%; Max-size prints 31%	Mixed background 48%; Max-size prints 33%

Strike-specific findings

- IV rule: corr(side, IV residual) -0.87 to -0.86; 94-95% of prints on the rule's side across days.
- Rule-consistent prints: 100-tick markout -0.06 to +1.19; other prints -4.47 to +1.15.

VOUCHER_7300 (position limit 50)

Players: Market-maker bot + IV-smile value taker (most prints) + residual noise taker. **Implication:** Avoid being the passive side to rule-consistent flow; skew quotes by the IV residual (see the vouchers family page)



Findings by day

- **Day -2:** 637 prints / 1272 lots, buys 54% (p = 0.03); arrival dispersion 2.79, sign ACF(1) +0.80; side is value-leaning (corr with EWMA deviation -0.45); aggressor markout -1.4 at 1 tick and +2.8 at 100 ticks vs half spread 1.5; dominant cluster 'Mixed background'.
- **Day -1:** 587 prints / 1186 lots, buys 58% (p = 0.00); arrival dispersion 3.84, sign ACF(1) +0.83; side is value-leaning (corr with EWMA deviation -0.39); aggressor markout -1.5 at 1 tick and +1.0 at 100 ticks vs half spread 1.5; dominant cluster 'Mixed background'.
- **Day 0:** 639 prints / 1287 lots, buys 54% (p = 0.08); arrival dispersion 3.01, sign ACF(1) +0.80; side is value-leaning (corr with EWMA deviation -0.26); aggressor markout -1.5 at 1 tick and +2.2 at 100 ticks vs half spread 1.5; dominant cluster 'Mixed background'.

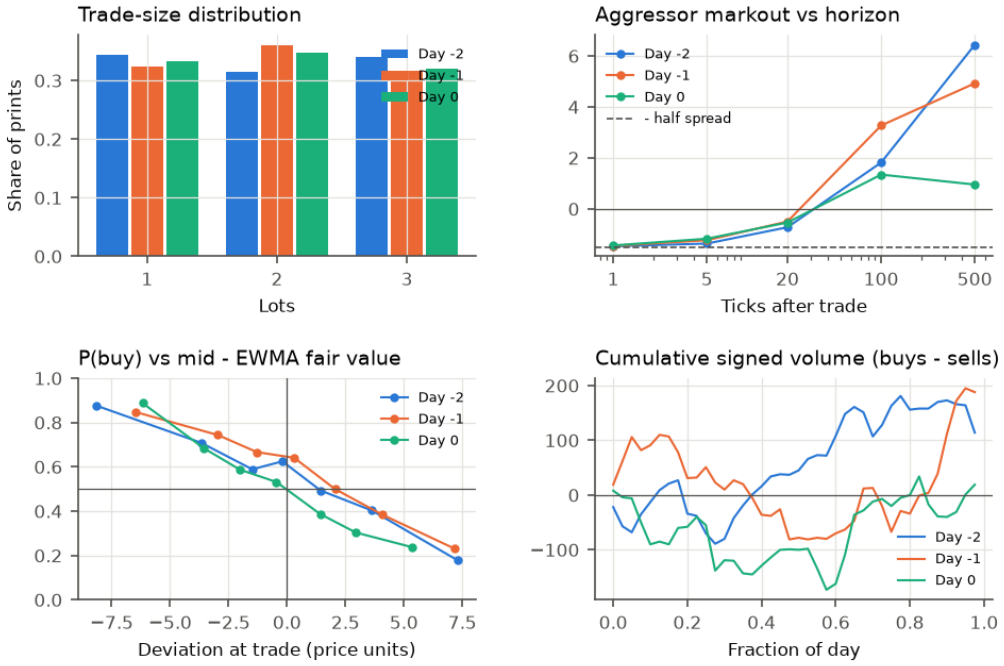
Metric	Day -2	Day -1	Day 0
Prints / lots (avg size)	637 / 1272 (2.00)	587 / 1186 (2.02)	639 / 1287 (2.01)
Size range / uniform p	1-3 / 0.851	1-3 / 0.765	1-3 / 0.906
L2+ sweeps / multi-print ts	0% / 0%	0% / 0%	0% / 0%
Lots traded / position limit	25.4x	23.7x	25.7x
Count dispersion / gap CV	2.79 / 1.53	3.84 / 2.05	3.01 / 1.78
Buy share (binomial p)	54% (0.03)	58% (0.00)	54% (0.08)
Sign ACF(1) / runs ratio	+0.80 / 0.21	+0.83 / 0.17	+0.80 / 0.20
corr(side, EWMA dev / mom / range)	-0.45 / -0.23 / -0.58	-0.39 / -0.16 / -0.30	-0.26 / -0.15 / -0.48
Markout h1 / h20 / h100 / h500	-1.4 / -0.5 / +2.8 / +6.0	-1.5 / -0.8 / +1.0 / +5.9	-1.5 / -0.7 / +2.2 / +6.7
Eff. half spread / realized / impact (100)	1.50 / -2.80 / +4.30	1.50 / -1.02 / +2.52	1.50 / -2.20 / +3.70
Hasbrouck LR impact (half spreads)	+0.24	+0.05	+0.01
PIN (EKOP)	0.81	0.79	0.79
Top archetype clusters	Mixed background 34%; Max-size prints 33%	Mixed background 39%; Max-size prints 35%	Mixed background 41%; Max-size prints 34%

Strike-specific findings

- IV rule: corr(side, IV residual) -0.86 to -0.84; 93-95% of prints on the rule's side across days.
- Rule-consistent prints: 100-tick markout +1.20 to +3.17; other prints -3.50 to -1.35.

VOUCHER_7400 (position limit 50)

Players: Market-maker bot + IV-smile value taker (most prints) + residual noise taker. **Implication:** Avoid being the passive side to rule-consistent flow; skew quotes by the IV residual (see the vouchers family page)



Findings by day

- **Day -2:** 508 prints / 1014 lots, buys 55% (p = 0.02); arrival dispersion 3.24, sign ACF(1) +0.70; side is value-leaning (corr with EWMA deviation -0.42); aggressor markout -1.5 at 1 tick and +1.8 at 100 ticks vs half spread 1.5; dominant cluster 'Max-size prints'.
- **Day -1:** 547 prints / 1090 lots, buys 57% (p = 0.00); arrival dispersion 3.85, sign ACF(1) +0.77; side is value-leaning (corr with EWMA deviation -0.40); aggressor markout -1.5 at 1 tick and +3.3 at 100 ticks vs half spread 1.5; dominant cluster 'Mixed background'.
- **Day 0:** 557 prints / 1107 lots, buys 52% (p = 0.45); arrival dispersion 4.31, sign ACF(1) +0.70; side is value-leaning (corr with EWMA deviation -0.42); aggressor markout -1.4 at 1 tick and +1.3 at 100 ticks vs half spread 1.5; dominant cluster 'Mixed background'.

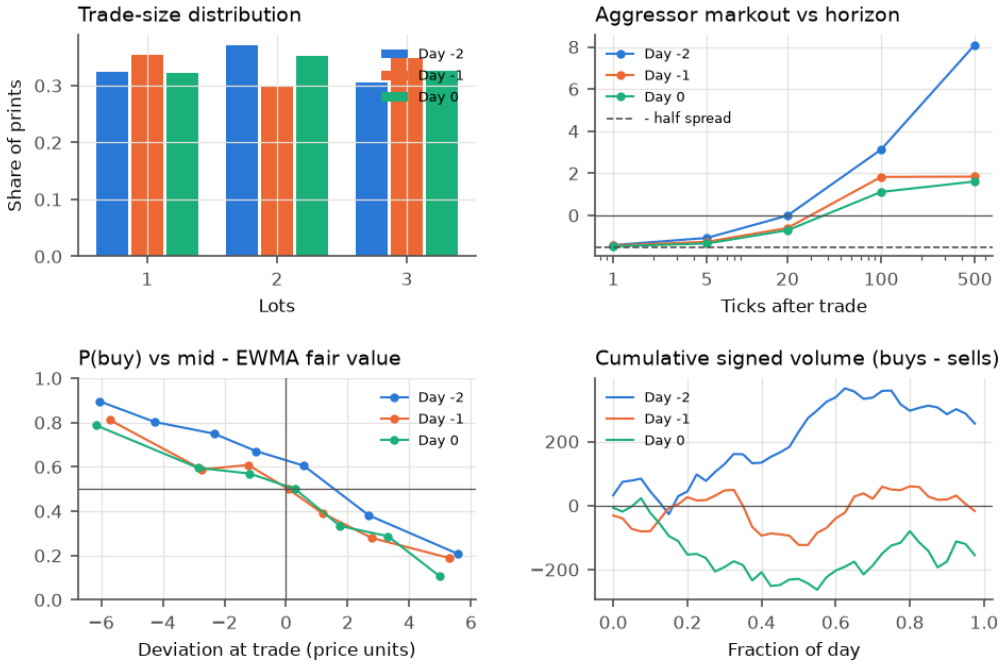
Metric	Day -2	Day -1	Day 0
Prints / lots (avg size)	508 / 1014 (2.00)	547 / 1090 (1.99)	557 / 1107 (1.99)
Size range / uniform p	1-3 / 0.676	1-3 / 0.404	1-3 / 0.707
L2+ sweeps / multi-print ts	0% / 0%	0% / 0%	0% / 0%
Lots traded / position limit	20.3x	21.8x	22.1x
Count dispersion / gap CV	3.24 / 1.88	3.85 / 1.80	4.31 / 1.75
Buy share (binomial p)	55% (0.02)	57% (0.00)	52% (0.45)
Sign ACF(1) / runs ratio	+0.70 / 0.31	+0.77 / 0.23	+0.70 / 0.30
corr(side, EWMA dev / mom / range)	-0.42 / -0.25 / -0.31	-0.40 / -0.23 / -0.29	-0.42 / -0.16 / -0.37
Markout h1 / h20 / h100 / h500	-1.5 / -0.7 / +1.8 / +6.4	-1.5 / -0.5 / +3.3 / +4.9	-1.4 / -0.5 / +1.3 / +1.0
Eff. half spread / realized / impact (100)	1.50 / -1.82 / +3.32	1.50 / -3.27 / +4.77	1.50 / -1.35 / +2.85
Hasbrouck LR impact (half spreads)	+0.06	+0.11	+0.15
PIN (EKOP)	0.71	0.63	0.64
Top archetype clusters	Max-size prints 34%; Mixed background 32%	Mixed background 37%; Max-size prints 32%	Mixed background 46%; Max-size prints 32%

Strike-specific findings

- IV rule: corr(side, IV residual) -0.82 to -0.78; 90-94% of prints on the rule's side across days.
- Rule-consistent prints: 100-tick markout +1.90 to +3.46; other prints -3.90 to +0.29.

VOUCHER_7500 (position limit 50)

Players: Market-maker bot + IV-smile value taker (most prints) + residual noise taker. **Implication:** Avoid being the passive side to rule-consistent flow; skew quotes by the IV residual (see the vouchers family page)



Findings by day

- **Day -2:** 534 prints / 1058 lots, buys 62% (p = 0.00); arrival dispersion 2.59, sign ACF(1) +0.78; side is value-leaning (corr with EWMA deviation -0.45); aggressor markout -1.4 at 1 tick and +3.1 at 100 ticks vs half spread 1.5; dominant cluster 'Mixed background'.
- **Day -1:** 480 prints / 957 lots, buys 48% (p = 0.44); arrival dispersion 3.86, sign ACF(1) +0.78; side is value-leaning (corr with EWMA deviation -0.40); aggressor markout -1.4 at 1 tick and +1.8 at 100 ticks vs half spread 1.5; dominant cluster 'Max-size prints'.
- **Day 0:** 656 prints / 1315 lots, buys 45% (p = 0.02); arrival dispersion 2.75, sign ACF(1) +0.78; side is value-leaning (corr with EWMA deviation -0.42); aggressor markout -1.5 at 1 tick and +1.1 at 100 ticks vs half spread 1.5; dominant cluster 'Mixed background'.

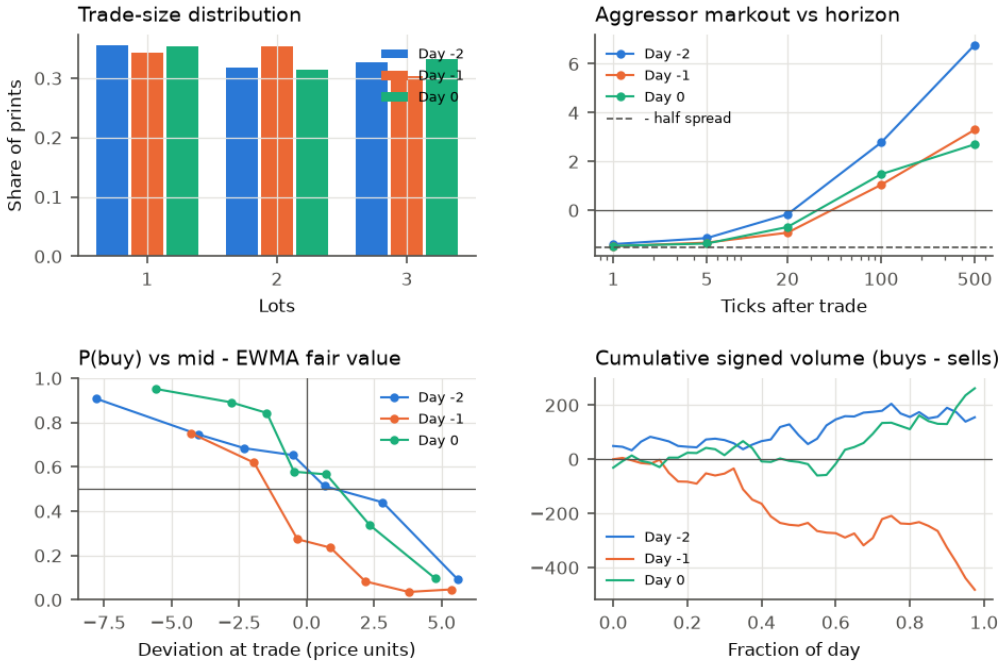
Metric	Day -2	Day -1	Day 0
Prints / lots (avg size)	534 / 1058 (1.98)	480 / 957 (1.99)	656 / 1315 (2.00)
Size range / uniform p	1-3 / 0.161	1-3 / 0.254	1-3 / 0.587
L2+ sweeps / multi-print ts	0% / 0%	0% / 0%	0% / 0%
Lots traded / position limit	21.2x	19.1x	26.3x
Count dispersion / gap CV	2.59 / 1.67	3.86 / 1.75	2.75 / 1.58
Buy share (binomial p)	62% (0.00)	48% (0.44)	45% (0.02)
Sign ACF(1) / runs ratio	+0.78 / 0.22	+0.78 / 0.22	+0.78 / 0.23
corr(side, EWMA dev / mom / range)	-0.45 / -0.27 / -0.15	-0.40 / -0.21 / -0.45	-0.42 / -0.24 / -0.46
Markout h1 / h20 / h100 / h500	-1.4 / -0.0 / +3.1 / +8.1	-1.4 / -0.6 / +1.8 / +1.8	-1.5 / -0.7 / +1.1 / +1.6
Eff. half spread / realized / impact (100)	1.50 / -3.13 / +4.63	1.50 / -1.83 / +3.33	1.50 / -1.12 / +2.62
Hasbrouck LR impact (half spreads)	+0.29	+0.12	+0.07
PIN (EKOP)	0.77	0.78	0.75
Top archetype clusters	Mixed background 42%; Max-size prints 31%	Max-size prints 35%; Mixed background 31%	Mixed background 38%; Max-size prints 33%

Strike-specific findings

- IV rule: corr(side, IV residual) -0.84 to -0.82; 91-93% of prints on the rule's side across days.
- Rule-consistent prints: 100-tick markout +1.43 to +3.53; other prints -3.24 to -2.39.

VOUCHER_7600 (position limit 50)

Players: Market-maker bot + IV-smile value taker (most prints) + residual noise taker. **Implication:** Avoid being the passive side to rule-consistent flow; skew quotes by the IV residual (see the vouchers family page)



Findings by day

- **Day -2:** 529 prints / 1043 lots, buys 58% (p = 0.00); arrival dispersion 2.86, sign ACF(1) +0.67; side is value-leaning (corr with EWMA deviation -0.48); aggressor markout -1.4 at 1 tick and +2.8 at 100 ticks vs half spread 1.5; dominant cluster 'Max-size prints'.
- **Day -1:** 591 prints / 1158 lots, buys 29% (p = 0.00); arrival dispersion 4.20, sign ACF(1) +0.75; side is value-leaning (corr with EWMA deviation -0.54); aggressor markout -1.5 at 1 tick and +1.0 at 100 ticks vs half spread 1.5; dominant cluster 'Contrarian / value'.
- **Day 0:** 581 prints / 1150 lots, buys 61% (p = 0.00); arrival dispersion 3.27, sign ACF(1) +0.74; side is value-leaning (corr with EWMA deviation -0.55); aggressor markout -1.5 at 1 tick and +1.5 at 100 ticks vs half spread 1.5; dominant cluster 'Mixed background'.

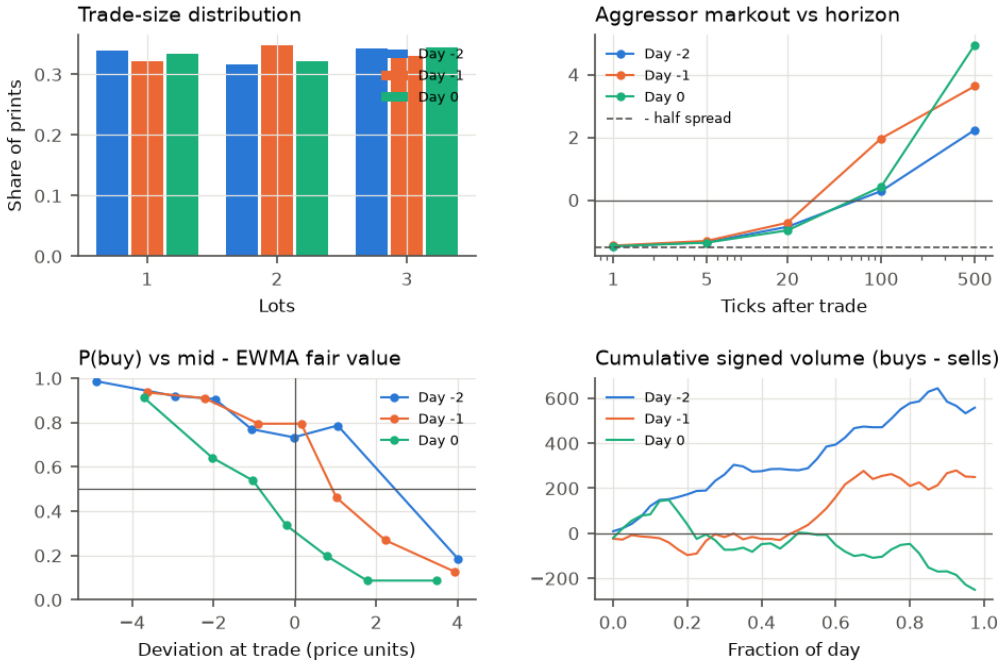
Metric	Day -2	Day -1	Day 0
Prints / lots (avg size)	529 / 1043 (1.97)	591 / 1158 (1.96)	581 / 1150 (1.98)
Size range / uniform p	1-3 / 0.541	1-3 / 0.278	1-3 / 0.534
L2+ sweeps / multi-print ts	0% / 0%	0% / 0%	0% / 0%
Lots traded / position limit	20.9x	23.2x	23.0x
Count dispersion / gap CV	2.86 / 1.56	4.20 / 1.81	3.27 / 1.65
Buy share (binomial p)	58% (0.00)	29% (0.00)	61% (0.00)
Sign ACF(1) / runs ratio	+0.67 / 0.33	+0.75 / 0.26	+0.74 / 0.26
corr(side, EWMA dev / mom / range)	-0.48 / -0.26 / -0.51	-0.54 / -0.34 / -0.64	-0.55 / -0.30 / -0.57
Markout h1 / h20 / h100 / h500	-1.4 / -0.2 / +2.8 / +6.8	-1.5 / -0.9 / +1.0 / +3.3	-1.5 / -0.7 / +1.5 / +2.7
Eff. half spread / realized / impact (100)	1.50 / -2.78 / +4.28	1.50 / -1.04 / +2.54	1.50 / -1.48 / +2.98
Hasbrouck LR impact (half spreads)	+0.19	+0.10	+0.06
PIN (EKOP)	0.62	0.57	0.67
Top archetype clusters	Max-size prints 33%; Contrarian / value 31%	Contrarian / value 35%; Max-size prints 30%	Mixed background 35%; Max-size prints 33%

Strike-specific findings

- IV rule: corr(side, IV residual) -0.81 to -0.80; 91-93% of prints on the rule's side across days.
- Rule-consistent prints: 100-tick markout +1.43 to +3.25; other prints -2.89 to -1.22.

VOUCHER_7750 (position limit 50)

Players: Market-maker bot + IV-smile value taker (most prints) + residual noise taker. **Implication:** Avoid being the passive side to rule-consistent flow; skew quotes by the IV residual (see the vouchers family page)



Findings by day

- **Day -2:** 524 prints / 1050 lots, buys 76% (p = 0.00); arrival dispersion 3.47, sign ACF(1) +0.68; side is value-leaning (corr with EWMA deviation -0.53); aggressor markout -1.5 at 1 tick and +0.3 at 100 ticks vs half spread 1.5; dominant cluster 'Contrarian / value'.
- **Day -1:** 546 prints / 1096 lots, buys 61% (p = 0.00); arrival dispersion 2.73, sign ACF(1) +0.77; side is value-leaning (corr with EWMA deviation -0.57); aggressor markout -1.5 at 1 tick and +2.0 at 100 ticks vs half spread 1.5; dominant cluster 'Mixed background'.
- **Day 0:** 642 prints / 1291 lots, buys 40% (p = 0.00); arrival dispersion 2.85, sign ACF(1) +0.82; side is value-leaning (corr with EWMA deviation -0.55); aggressor markout -1.5 at 1 tick and +0.4 at 100 ticks vs half spread 1.5; dominant cluster 'Max-size prints'.

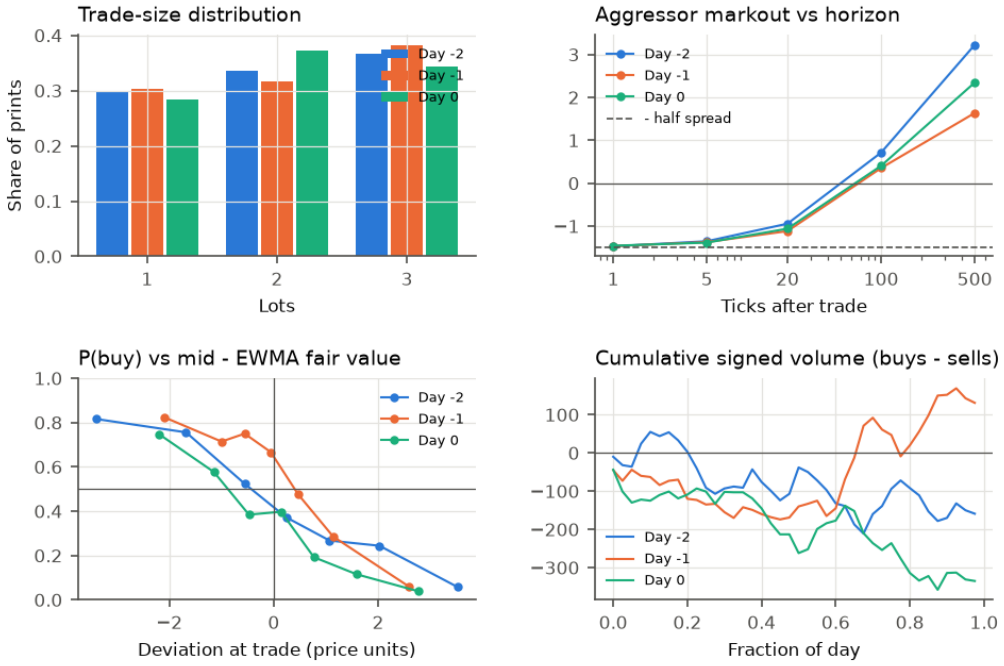
Metric	Day -2	Day -1	Day 0
Prints / lots (avg size)	524 / 1050 (2.00)	546 / 1096 (2.01)	642 / 1291 (2.01)
Size range / uniform p	1-3 / 0.720	1-3 / 0.751	1-3 / 0.795
L2+ sweeps / multi-print ts	0% / 0%	0% / 0%	0% / 0%
Lots traded / position limit	21.0x	21.9x	25.8x
Count dispersion / gap CV	3.47 / 1.91	2.73 / 1.91	2.85 / 1.70
Buy share (binomial p)	76% (0.00)	61% (0.00)	40% (0.00)
Sign ACF(1) / runs ratio	+0.68 / 0.32	+0.77 / 0.23	+0.82 / 0.18
corr(side, EWMA dev / mom / range)	-0.53 / -0.32 / -0.52	-0.57 / -0.42 / -0.57	-0.55 / -0.28 / -0.72
Markout h1 / h20 / h100 / h500	-1.5 / -0.9 / +0.3 / +2.2	-1.5 / -0.7 / +2.0 / +3.6	-1.5 / -1.0 / +0.4 / +4.9
Eff. half spread / realized / impact (100)	1.50 / -0.28 / +1.78	1.50 / -1.96 / +3.46	1.50 / -0.42 / +1.92
Hasbrouck LR impact (half spreads)	+0.06	+0.07	+0.08
PIN (EKOP)	0.77	0.72	0.78
Top archetype clusters	Contrarian / value 41%; Max-size prints 34%	Mixed background 38%; Max-size prints 33%	Max-size prints 34%; Mixed background 31%

Strike-specific findings

- IV rule: corr(side, IV residual) -0.86 to -0.76; 91-95% of prints on the rule's side across days.
- Rule-consistent prints: 100-tick markout +0.56 to +2.24; other prints -2.58 to -1.38.

VOUCHER_7900 (position limit 50)

Players: Market-maker bot + IV-smile value taker (most prints) + residual noise taker. **Implication:** Avoid being the passive side to rule-consistent flow; skew quotes by the IV residual (see the vouchers family page)



Findings by day

- **Day -2:** 604 prints / 1249 lots, buys 43% (p = 0.00); arrival dispersion 3.00, sign ACF(1) +0.82; side is value-leaning (corr with EWMA deviation -0.51); aggressor markout -1.5 at 1 tick and +0.7 at 100 ticks vs half spread 1.5; dominant cluster 'Max-size prints'.
- **Day -1:** 589 prints / 1225 lots, buys 54% (p = 0.06); arrival dispersion 3.72, sign ACF(1) +0.78; side is value-leaning (corr with EWMA deviation -0.51); aggressor markout -1.5 at 1 tick and +0.4 at 100 ticks vs half spread 1.5; dominant cluster 'Mixed background'.
- **Day 0:** 547 prints / 1127 lots, buys 35% (p = 0.00); arrival dispersion 3.43, sign ACF(1) +0.78; side is value-leaning (corr with EWMA deviation -0.50); aggressor markout -1.5 at 1 tick and +0.4 at 100 ticks vs half spread 1.5; dominant cluster 'Mixed background'.

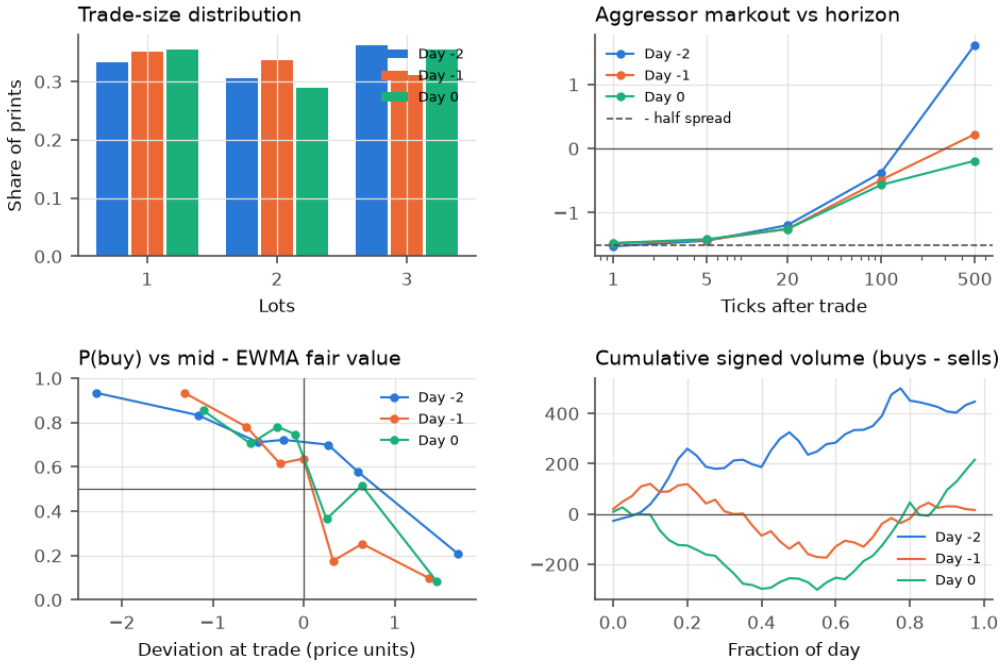
Metric	Day -2	Day -1	Day 0
Prints / lots (avg size)	604 / 1249 (2.07)	589 / 1225 (2.08)	547 / 1127 (2.06)
Size range / uniform p	1-3 / 0.123	1-3 / 0.040	1-3 / 0.033
L2+ sweeps / multi-print ts	0% / 0%	0% / 0%	0% / 0%
Lots traded / position limit	25.0x	24.5x	22.5x
Count dispersion / gap CV	3.00 / 1.74	3.72 / 1.73	3.43 / 1.72
Buy share (binomial p)	43% (0.00)	54% (0.06)	35% (0.00)
Sign ACF(1) / runs ratio	+0.82 / 0.18	+0.78 / 0.22	+0.78 / 0.23
corr(side, EWMA dev / mom / range)	-0.51 / -0.32 / -0.58	-0.51 / -0.30 / -0.39	-0.50 / -0.30 / -0.63
Markout h1 / h20 / h100 / h500	-1.5 / -0.9 / +0.7 / +3.2	-1.5 / -1.1 / +0.4 / +1.6	-1.5 / -1.1 / +0.4 / +2.3
Eff. half spread / realized / impact (100)	1.50 / -0.71 / +2.21	1.50 / -0.35 / +1.85	1.50 / -0.41 / +1.91
Hasbrouck LR impact (half spreads)	+0.08	+0.06	+0.05
PIN (EKOP)	0.85	0.74	0.76
Top archetype clusters	Max-size prints 37%; Mixed background 31%	Mixed background 40%; Max-size prints 38%	Mixed background 36%; Max-size prints 34%

Strike-specific findings

- IV rule: corr(side, IV residual) -0.83 to -0.75; 86-92% of prints on the rule's side across days.
- Rule-consistent prints: 100-tick markout +0.51 to +0.84; other prints -1.21 to -0.70.

VOUCHER_8100 (position limit 50)

Players: Market-maker bot + IV-smile value taker (most prints) + residual noise taker. **Implication:** Avoid being the passive side to rule-consistent flow; skew quotes by the IV residual (see the vouchers family page)



Findings by day

- **Day -2:** 632 prints / 1283 lots, buys 67% (p = 0.00); arrival dispersion 3.63, sign ACF(1) +0.82; side is value-leaning (corr with EWMA deviation -0.45); aggressor markout -1.5 at 1 tick and -0.4 at 100 ticks vs half spread 1.5; dominant cluster 'Contrarian / value'.
- **Day -1:** 637 prints / 1248 lots, buys 50% (p = 1.00); arrival dispersion 2.79, sign ACF(1) +0.78; side is value-leaning (corr with EWMA deviation -0.53); aggressor markout -1.5 at 1 tick and -0.5 at 100 ticks vs half spread 1.5; dominant cluster 'Mixed background'.
- **Day 0:** 669 prints / 1338 lots, buys 58% (p = 0.00); arrival dispersion 3.99, sign ACF(1) +0.83; side is value-leaning (corr with EWMA deviation -0.46); aggressor markout -1.5 at 1 tick and -0.6 at 100 ticks vs half spread 1.5; dominant cluster 'Max-size prints'.

Metric	Day -2	Day -1	Day 0
Prints / lots (avg size)	632 / 1283 (2.03)	637 / 1248 (1.96)	669 / 1338 (2.00)
Size range / uniform p	1-3 / 0.214	1-3 / 0.440	1-3 / 0.048
L2+ sweeps / multi-print ts	0% / 0%	0% / 0%	0% / 0%
Lots traded / position limit	25.7x	25.0x	26.8x
Count dispersion / gap CV	3.63 / 1.75	2.79 / 1.66	3.99 / 1.86
Buy share (binomial p)	67% (0.00)	50% (1.00)	58% (0.00)
Sign ACF(1) / runs ratio	+0.82 / 0.18	+0.78 / 0.22	+0.83 / 0.18
corr(side, EWMA dev / mom / range)	-0.45 / -0.23 / -0.68	-0.53 / -0.28 / -0.69	-0.46 / -0.25 / -0.81
Markout h1 / h20 / h100 / h500	-1.5 / -1.2 / -0.4 / +1.6	-1.5 / -1.3 / -0.5 / +0.2	-1.5 / -1.3 / -0.6 / -0.2
Eff. half spread / realized / impact (100)	1.50 / +0.37 / +1.13	1.50 / +0.49 / +1.01	1.50 / +0.57 / +0.93
Hasbrouck LR impact (half spreads)	+0.03	+0.03	+0.02
PIN (EKOP)	0.76	0.72	0.82
Top archetype clusters	Contrarian / value 39%; Max-size prints 36%	Mixed background 31%; Max-size prints 31%	Max-size prints 36%; Contrarian / value 35%

Strike-specific findings

- IV rule: corr(side, IV residual) -0.71 to -0.58; 76-84% of prints on the rule's side across days.
- Rule-consistent prints: 100-tick markout -0.56 to -0.32; other prints -0.74 to -0.60.

Strategy implications

- **Make markets wherever the counterparty is the noise taker** (15 products): adverse selection is ~0, so spread capture is limited only by price noise and inventory.
- **Vouchers: trade with the IV rule, not against it.** Skew quotes so we are not the passive side to rule-consistent flow at 7,300-7,600; consider taking the same side first when the residual is large.
- **TELLURIUM: jump-aware quoting.** After a jump, pull the side the chasers will hit, or move it beyond L2, for ~200 ticks; re-enter as intensity decays.
- **PINK_PEARLS and BLUE_PEARLS signals are uncontested** by any observed taker, so execution competes only with the market-maker bot's quotes.

Caveats

- Counterparty IDs are blank; every archetype is a behavioural inference from three days.
- Thin products (12-33 prints/day in the basket family) give low statistical power; labels there default to noise.
- PIN for products with < 100 prints/day swings between 0 and 1 across days and should not be used.
- The IV residual depends on the 8/7/6-day expiry convention from the EDA report; the sign of the rule is robust to it, the magnitudes less so.
- Market trades in the CSVs are bot-vs-bot. Once our orders rest in the book, the QuantiFi passive-fill rule (not queue priority) decides which of these prints we would receive.

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